

第五章 电磁波的传播

§ 5.1 平面电磁波

1. 自由空间

$$J_f = 0, \rho_f = 0$$

$$\begin{cases} \nabla \cdot \vec{D} = 0 \\ \nabla \times \vec{H} = \frac{\partial \vec{D}}{\partial t} \\ \nabla \cdot \vec{B} = 0 \\ \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \end{cases} \quad \text{其中} \quad \vec{D} = \epsilon \vec{E}, \vec{B} = \mu \vec{H} \Rightarrow \begin{cases} \nabla \cdot \vec{E} = 0 \\ \nabla \times \vec{B} = \mu \epsilon \frac{\partial \vec{E}}{\partial t} \\ \nabla \cdot \vec{B} = 0 \\ \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \end{cases}$$

$$\hookrightarrow \begin{cases} \nabla^2 \vec{E}(\vec{r}, t) - \mu \epsilon \frac{\partial^2 \vec{E}(\vec{r}, t)}{\partial t^2} = 0 \\ \nabla^2 \vec{B}(\vec{r}, t) - \mu \epsilon \frac{\partial^2 \vec{B}(\vec{r}, t)}{\partial t^2} = 0 \end{cases}$$

2. 时谐电磁波

$$\vec{E}(\vec{r}, t) = \vec{E}(\vec{r}) e^{-i\omega t}, \quad \vec{B}(\vec{r}, t) = \vec{B}(\vec{r}) e^{-i\omega t}$$

$$\hookrightarrow \begin{cases} (\nabla^2 + \omega^2 \mu \epsilon) \vec{E}(\vec{r}) = 0 & k = \omega \sqrt{\mu \epsilon} = \frac{\omega}{c} n = nk_0 \\ (\nabla^2 + \omega^2 \mu \epsilon) \vec{B}(\vec{r}) = 0 \end{cases}$$

$$\begin{cases} (\nabla^2 + k^2) \vec{E}(\vec{r}) = 0 & \text{波动方程} \\ (\nabla^2 + k^2) \vec{B}(\vec{r}) = 0 \end{cases}$$

$$\begin{cases} \vec{E}(\vec{r}) = -\frac{i\omega}{k^2} \nabla \times \vec{B}(\vec{r}) \\ \vec{B}(\vec{r}) = -\frac{i}{\omega} \nabla \times \vec{E}(\vec{r}) \end{cases}$$

3. 单色平面波.

$$\vec{E}(\vec{r}) = \vec{E}_0 e^{i\vec{k} \cdot \vec{r} - \omega t + \phi_0}$$

$$\text{相位: } \phi = \vec{k} \cdot \vec{r} - \omega t + \phi_0$$

$$\text{相速度 } \vec{v}_p = \frac{\partial \vec{r}}{\partial t} = v \vec{e}_k$$

$$\nabla \cdot \vec{E}(\vec{r}, t) = 0 \rightarrow \vec{e}_k \cdot \vec{E} = 0,$$

$$\nabla \cdot \vec{B}(\vec{r}, t) = 0 \rightarrow \vec{e}_k \cdot \vec{B} = 0$$

$\left. \begin{array}{l} \vec{E} \cdot \vec{B} \perp \vec{k}, \text{ TEM模.} \end{array} \right\}$

$$\vec{B}(\vec{r}) = \frac{1}{\omega} \nabla \times \vec{E}(\vec{r}) \quad \vec{E}, \vec{B} \text{ 同相位}$$

$$\vec{E}(\vec{r}) = v \vec{B}(\vec{r}) \times \vec{e}_k$$

$$\text{振幅: } \frac{|\vec{E}_0|}{|\vec{B}_0|} = v$$

$$\text{阻抗: } Z = \sqrt{\frac{\mu}{\epsilon}} = Z_0 \sqrt{\frac{\mu_r}{\epsilon_r}}$$

$$\frac{|\vec{E}_0|}{|\vec{H}_0|} = v \cdot \mu = \sqrt{\frac{\mu}{\epsilon}} = Z$$

4. 能量密度, 动量密度.

$$\begin{aligned} w(\vec{r}, t) &= \frac{1}{2} (\vec{E} \cdot \vec{D} + \vec{H} \cdot \vec{B}) = \frac{1}{2} (\epsilon \vec{E}^2 + \frac{1}{\mu} \vec{B}^2) = \epsilon(\vec{r}) [\vec{E}(\vec{r}, t)]^2 \\ &= \epsilon |\vec{E}_0|^2 \cos^2(\vec{k} \cdot \vec{r} - \omega t + \phi_0) \end{aligned}$$

$$\langle w \rangle = \frac{1}{2} \epsilon |\vec{E}_0|^2$$

$$\vec{S} = \vec{E} \times \vec{H} = \frac{1}{\mu} (\vec{E} \times \vec{B})$$

$$= \frac{1}{\mu} \vec{E} \times \frac{1}{v} (\vec{e}_k \times \vec{E}) = \frac{\epsilon}{\mu} E^2 \vec{e}_k$$

$$\langle \vec{S} \rangle = \frac{1}{2} \frac{\epsilon}{\mu} \langle |\vec{E}_0|^2 \rangle \vec{e}_k = \frac{\epsilon}{2\mu} \langle \frac{1}{2} \epsilon_0 |\vec{E}_0|^2 \rangle \vec{e}_k$$

$$= v_p \langle w \rangle \vec{e}_k$$

$$I = |\langle \vec{S} \rangle| = v_p \langle w \rangle$$

$$v_g = \frac{\langle \vec{S} \rangle}{\langle w \rangle} = v_p \vec{e}_k$$

5. 动量密度, 动量流密度.

$$\text{动量密度 } \vec{g} = \epsilon \vec{E} \times \left(\frac{1}{v} \vec{e}_k \times \vec{E} \right) = \frac{\epsilon}{v_p} \vec{E}^2 \vec{e}_k = \frac{w}{v_p} \vec{e}_k$$

$$\langle \vec{g} \rangle = \frac{\langle w \rangle}{v_p} \vec{e}_k$$

$$\text{动量流密度 } \vec{T} = - \left\{ \vec{B} \vec{E} + \vec{B} \vec{H} - \frac{1}{2} (\vec{B} \cdot \vec{E} + \vec{B} \cdot \vec{H}) \vec{I} \right\}$$

$$T_{ij} = - (D_i E_j + B_i H_j) + \frac{1}{2} (\vec{D} \cdot \vec{E} + \vec{B} \cdot \vec{H}) \delta_{ij}$$

考虑沿 \$z\$ 轴传播, \$\vec{E} \propto \vec{e}_x\$, \$\vec{B} \propto \vec{e}_y\$

$$T = \frac{1}{2} \begin{pmatrix} -\epsilon E_x^2 + \mu H_y^2 & 0 & 0 \\ 0 & \epsilon E_x^2 - \mu H_y^2 & 0 \\ 0 & 0 & \epsilon E_z^2 + \mu H_y^2 \end{pmatrix}$$

$$\epsilon \vec{E}^2 = \mu \vec{H}^2$$

$$\hookrightarrow T_{11} = T_{22} = 0, \quad T_{33} = \text{Re}(\epsilon E_z^2), \quad T_{ij} = 0$$

$$\langle \vec{T} \rangle = \langle T_{33} \rangle \vec{e}_z \vec{e}_z = \frac{1}{2} \epsilon |\vec{E}_0|^2$$

例: 计算单色平面波入射到理想导体表面所产生辐射压强.

$$\vec{G} = \vec{g} \Delta V = g \cos \theta \Delta S \cos \theta$$

$$P = \frac{2G \cos \theta}{\Delta S \cos \theta} = 2g \cos^2 \theta = 2 \langle \omega \rangle \cos^2 \theta = \frac{\epsilon E^2}{2} \cos^2 \theta$$

6. 单色平面波的偏振椭圆.

选取基底 \$\{\vec{e}_1, \vec{e}_2\}\$ \$\vec{e}_1 \times \vec{e}_2 = \vec{e}_z\$

$$\vec{E}(\vec{R}, t) = (E_{10} \vec{e}_1 + E_{20} \vec{e}_2) e^{i\vec{k} \cdot \vec{R} - i\omega t} \quad E_{10} = |E_{10}| e^{i\phi_1}, \quad E_{20} = |E_{20}| e^{i\phi_2}$$

$$\begin{aligned} \text{Re}(\vec{E}) &= |E_{10}| \cos(\phi + \delta_1) \vec{e}_1 + |E_{20}| \cos(\phi + \delta_2) \vec{e}_2 \\ &= E_1 \vec{e}_1 + E_2 \vec{e}_2 \end{aligned}$$

$$\text{其中 } \begin{cases} \frac{E_1}{|E_{10}|} = \cos \phi \cos \delta_1 - \sin \phi \sin \delta_1 \\ \frac{E_2}{|E_{20}|} = \cos \phi \cos \delta_2 - \sin \phi \sin \delta_2 \end{cases}$$

$$\hookrightarrow \begin{cases} \frac{E_1}{|E_{10}|} \sin \delta_2 - \frac{E_2}{|E_{20}|} \sin \delta_1 = \cos \phi \sin(\delta_1 - \delta_2) \\ \frac{E_1}{|E_{10}|} \cos \delta_2 - \frac{E_2}{|E_{20}|} \cos \delta_1 = \sin \phi \sin(\delta_2 - \delta_1) \end{cases}$$

$$\Rightarrow \left(\frac{E_1}{|E_{10}|} \right)^2 + \left(\frac{E_2}{|E_{20}|} \right)^2 - 2 \frac{E_1}{|E_{10}|} \frac{E_2}{|E_{20}|} \cos \delta = \sin^2 \delta$$

$$\delta = \delta_2 - \delta_1$$

7. 线偏振.

$$\delta = \delta_2 - \delta_1 = m\pi$$

$$\text{Re}(\vec{E}) = (|E_{10}| \vec{e}_1 + |E_{20}| \vec{e}_2) \cos(\vec{k} \cdot \vec{r} - \omega t + \delta_1)$$

8. 圆偏振.

$$\begin{cases} |E_{10}| = |E_{20}| = \frac{E_0}{\sqrt{2}} \\ \delta_1 - \delta_2 = \frac{1}{2} m\pi \end{cases}$$

$$\begin{aligned} \text{Re}(\vec{E}) &= |E_{10}| \cos(\vec{k} \cdot \vec{r} - \omega t) \vec{e}_1 + |E_{20}| \cos(\vec{k} \cdot \vec{r} - \omega t + \delta) \vec{e}_2 \\ &= \frac{E_0}{\sqrt{2}} \left[\cos(\vec{k} \cdot \vec{r} - \omega t) \vec{e}_1 + \cos(\vec{k} \cdot \vec{r} - \omega t + \delta) \vec{e}_2 \right] \end{aligned}$$

$$\begin{aligned} \text{Re}[\vec{E}(\omega, t)] &= \frac{E_0}{\sqrt{2}} \left[\cos(-\omega t) \vec{e}_1 + \cos(-\omega t + \delta) \vec{e}_2 \right] \\ &= \frac{E_0}{\sqrt{2}} \left[\cos(\omega t) \vec{e}_1 + \sin(\omega t) \sin(\frac{m}{2}\pi) \vec{e}_2 \right] \end{aligned}$$

$$\begin{cases} \text{Re}[\vec{E}_{\text{LCP}}(\omega, t)] = \frac{E_0}{\sqrt{2}} \left[\cos(\omega t) \vec{e}_1 + \sin(\omega t) \sin(\frac{m}{2}\pi) \vec{e}_2 \right] \\ \text{Re}[\vec{E}_{\text{RCP}}(\omega, t)] = \frac{E_0}{\sqrt{2}} \left[\cos(\omega t) \vec{e}_1 - \sin(\omega t) \sin(\frac{m}{2}\pi) \vec{e}_2 \right] \end{cases}$$

$$\begin{cases} \vec{E}_{\text{LCP}}(\vec{r}, t) = \frac{E_0}{\sqrt{2}} (\vec{e}_1 + i\vec{e}_2) e^{i(\vec{k} \cdot \vec{r} - \omega t)} \\ \vec{E}_{\text{RCP}}(\vec{r}, t) = \frac{E_0}{\sqrt{2}} (\vec{e}_1 - i\vec{e}_2) e^{i(\vec{k} \cdot \vec{r} - \omega t)} \end{cases}$$

$$\vec{e}_+ = \frac{1}{\sqrt{2}} (\vec{e}_1 + i\vec{e}_2), \quad \vec{e}_- = \frac{1}{\sqrt{2}} (\vec{e}_1 - i\vec{e}_2)$$

$$\begin{cases} \vec{E}_{\text{LCP}}(\vec{r}, t) = E_0 \vec{e}_+ e^{i(\vec{k} \cdot \vec{r} - \omega t)} \\ \vec{E}_{\text{RCP}}(\vec{r}, t) = E_0 \vec{e}_- e^{i(\vec{k} \cdot \vec{r} - \omega t)} \end{cases}$$

方向: $\vec{E}(\vec{r}, t) = (\vec{e}_+ \vec{E}_{\text{LCP}} + \vec{e}_- \vec{E}_{\text{RCP}}) e^{i(\vec{k} \cdot \vec{r} - \omega t)}$

归一化: $\vec{e}_+ \cdot \vec{e}_+^* = \vec{e}_- \cdot \vec{e}_-^* = 1$

$$\vec{e}_- \cdot \vec{e}_+^* = \vec{e}_+ \cdot \vec{e}_-^* = 0$$

$$\vec{E} = \vec{E}_+ \vec{e}_+ + \vec{E}_- \vec{e}_-$$

$$= (\vec{E} \cdot \vec{e}_+^*) \vec{e}_+ + (\vec{E} \cdot \vec{e}_-^*) \vec{e}_-$$

§ 5.2 电磁波在介质分界面上的反射和折射

$$\left\{ \begin{array}{l} \text{入射场: } \vec{E}(\vec{r}, t) = \vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \\ \text{反射场: } \vec{E}'(\vec{r}, t) = \vec{E}'_0 e^{i(\vec{k}' \cdot \vec{r} - \omega t)} \\ \text{折射场: } \vec{E}''(\vec{r}, t) = \vec{E}''_0 e^{i(\vec{k}'' \cdot \vec{r} - \omega t)} \end{array} \right.$$

$$k_1 = k_1' = n_1 k_0, \quad k_2'' = n_2 k_0 \quad n = \sqrt{\epsilon_r \mu_r}$$

$$\left\{ \begin{array}{l} E_{1t} = E_{1t}, \quad H_{1t} = H_{1t} \\ D_{2n} = D_{1n}, \quad B_{2n} = B_{1n} \end{array} \right.$$

$$E_{1t} = \vec{n} \times \left[\vec{E}_0 e^{i(\vec{k}_1 \cdot \vec{r} - \omega t)} + \vec{E}'_0 e^{i(\vec{k}'_1 \cdot \vec{r} - \omega t)} \right] \Big|_{z=0}$$

$$E_{2t} = \vec{n} \times \vec{E}''_0 e^{i(\vec{k}'' \cdot \vec{r} - \omega t)} \Big|_{z=0}$$

$$\hookrightarrow k_{1x} = k_{1x}' = k_{2x}'', \quad k_{1y} = k_{1y}' = k_{2y}'' = 0 \quad (\text{假设入射面为 } xOz)$$

$$\text{代入 } k_{1x} = k_1 \sin \theta, \quad k_{1x}' = k_1' \sin \theta', \quad k_{2x}'' = k_2'' \sin \theta''$$

$$\hookrightarrow \left\{ \begin{array}{l} \theta = \theta' \quad \text{反射定律} \\ n_1 \sin \theta = n_2 \sin \theta'' \quad \text{折射定律} \end{array} \right.$$

2. Fresnel 公式

s 偏振: 电场垂直于入射面. TE

p 偏振: 磁场垂直于入射面. TM

(1) s 偏振

$$\begin{cases} E_0 + E_0' = E_0'' \\ H_0 \cos \theta - H_0' \cos \theta = H_0'' \cos \theta \end{cases}$$

$$Z = \frac{E}{H}$$

$$\textcircled{2} \rightarrow \frac{1}{Z_1} (E_0 \cos \theta - E_0' \cos \theta) = \frac{1}{Z_2} E_0'' \cos \theta''$$

$$\text{解得} \begin{cases} \left(\frac{E_0'}{E_0}\right)_s = \frac{Z_2 \cos \theta - Z_1 \cos \theta''}{Z_2 \cos \theta + Z_1 \cos \theta''} \\ \left(\frac{E_0''}{E_0}\right)_s = \frac{2Z_2 \cos \theta}{Z_2 \cos \theta + Z_1 \cos \theta''} \end{cases}$$

(2) p 偏振

$$\begin{cases} H_0 + H_0' = H_0'' \\ E_0 \cos \theta - E_0' \cos \theta'' = E_0'' \cos \theta'' \end{cases}$$

$$\rightarrow \frac{1}{Z_2} (-H_0 \cos \theta + H_0' \cos \theta) = -\frac{1}{Z_1} H_0'' \cos \theta''$$

$$\text{解得} \left(\frac{H_0'}{H_0}\right)_p = \frac{Z_1 \cos \theta - Z_2 \cos \theta''}{Z_1 \cos \theta + Z_2 \cos \theta''} \rightarrow \left(\frac{E_0'}{E_0}\right)_p = \frac{Z_1 \cos \theta - Z_2 \cos \theta''}{Z_1 \cos \theta + Z_2 \cos \theta''}$$

$$\left(\frac{H_0''}{H_0}\right)_p = \frac{2Z_1 \cos \theta}{Z_1 \cos \theta + Z_2 \cos \theta''} \rightarrow \left(\frac{E_0''}{E_0}\right)_p = \frac{2Z_2 \cos \theta}{Z_1 \cos \theta + Z_2 \cos \theta''}$$

(3) $\theta = 0$ (两种偏振相同)

$$\frac{E_0'}{E_0} = \frac{Z_2 - Z_1}{Z_2 + Z_1}, \quad \frac{E_0''}{E_0} = \frac{2Z_2}{Z_2 + Z_1}$$

阻抗匹配: $Z_1 = Z_2 \rightarrow E_0' = 0$ 无反射.

(4) 理想导体

$$|Z_2| \rightarrow \infty, \quad Z_2 = \sqrt{\frac{\mu_0}{\epsilon_0}} \rightarrow \infty$$

解得 $E_0' = -E_0$ 排斥电磁场, 完全反射.

3. 光波在介质分界面上的反射和折射。

在光学波段: $n \approx n_0$

$$\Rightarrow \sqrt{\epsilon_1} \sin \theta = \sqrt{\epsilon_2} \sin \theta'' \rightarrow \epsilon_2 \sin \theta = \epsilon_1 \sin \theta'' \quad \epsilon = \sqrt{\epsilon_0} \epsilon_r$$

$$\text{得 } \left(\frac{E_0'}{E_0} \right)_s = \frac{\sin(\theta'' - \theta)}{\sin(\theta'' + \theta)}, \quad \left(\frac{E_0'}{E_0} \right)_p = \frac{\tan(\theta'' - \theta)}{\tan(\theta'' + \theta)}$$

$$\left(\frac{E_0'}{E_0} \right)_s = \frac{2 \cos \theta \sin \theta''}{\sin(\theta'' + \theta)}, \quad \left(\frac{E_0'}{E_0} \right)_p = \frac{2 \cos \theta \sin \theta''}{\sin(\theta'' + \theta) \cos(\theta - \theta')}$$

4. 介质分界面的反射和透射系数。

$$\text{平均能流密度: } \langle S \rangle = \frac{1}{2} \sqrt{\frac{\epsilon_0}{\mu_0}} |E_0|^2 \vec{e}_k = \frac{1}{2} \frac{|E_0|^2}{Z} \vec{e}_k$$

$$\text{反射能流密度 } \langle S' \rangle = \frac{1}{2} \frac{|E_0'|^2}{Z_1} \vec{e}_k'$$

$$\text{折射能流密度 } \langle S \rangle = \frac{1}{2} \frac{|E_0|^2}{Z_2} \vec{e}_k$$

$$\text{反射系数 } R = \frac{\langle S' \rangle \cdot \vec{n}}{\langle S \rangle \cdot \vec{n}} = \frac{|E_0'|^2}{|E_0|^2} \frac{\vec{e}_k' \cdot \vec{n}}{\vec{e}_k \cdot \vec{n}} = \frac{|E_0'|^2}{|E_0|^2}$$

$$\text{透射系数 } T = \frac{\langle S \rangle \cdot \vec{n}}{\langle S' \rangle \cdot \vec{n}} = \frac{Z_1}{Z_2} \frac{|E_0|^2}{|E_0'|^2} \frac{\cos \theta}{\cos \theta''}$$

$$R_p = \left(\frac{E_0'}{E_0} \right)_p^2 = \left(\frac{Z_1 \cos \theta - Z_2 \cos \theta''}{Z_1 \cos \theta + Z_2 \cos \theta''} \right)^2, \quad R_s = \left(\frac{Z_2 \cos \theta - Z_1 \cos \theta''}{Z_2 \cos \theta + Z_1 \cos \theta''} \right)^2$$

$$T_p = \frac{4 Z_1 Z_2 \cos \theta \cos \theta''}{(Z_1 \cos \theta + Z_2 \cos \theta'')^2}, \quad T_s = \frac{4 Z_1 Z_2 \cos \theta \cos \theta''}{(Z_2 \cos \theta + Z_1 \cos \theta'')^2}$$

$$\text{垂直入射时: } R_p = R_s = \left(\frac{Z_1 - Z_2}{Z_1 + Z_2} \right)^2$$

$$T_p = T_s = \frac{4 Z_1 Z_2}{(Z_1 + Z_2)^2}$$

5. 光压

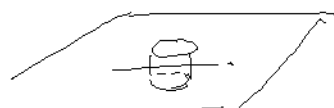
$$\vec{E} = E \vec{e}_x, \vec{H} = H \vec{e}_y$$

$$\text{动量流密度: } \vec{T} = - \left[\vec{D} \vec{E} + \vec{B} \vec{H} - \frac{1}{2} (\vec{D} \cdot \vec{E} + \vec{B} \cdot \vec{H}) \vec{I} \right]$$

$$\langle T_{ij} \rangle = \begin{pmatrix} -\epsilon E_x^2 + \mu_0 H_y^2 & \epsilon E_x E_y & \epsilon E_x^2 + \mu_0 H_y^2 \\ \epsilon E_x E_y & \epsilon E_x^2 - \mu_0 H_y^2 & \epsilon E_x E_y \\ \epsilon E_x^2 + \mu_0 H_y^2 & \epsilon E_x E_y & \epsilon E_x^2 + \mu_0 H_y^2 \end{pmatrix}$$

$$\text{光压 } P_{\text{rad}} = \frac{1}{\Delta S} \oint d\vec{S} \cdot \langle \vec{T} \rangle = \vec{e}_z \cdot \left[-\langle \vec{T} \rangle|_{z=0^-} + \langle \vec{T} \rangle|_{z=0^+} \right]$$

$$\left\{ \begin{array}{l} \vec{E}_1|_{z=0^+} = (E_0 + E_0') e^{-i\omega t} \vec{e}_x \\ H_1|_{z=0^+} = (H_0 + H_0') e^{-i\omega t} \vec{e}_y = \frac{1}{Z_1} (-E_0 + E_0') e^{-i\omega t} \vec{e}_y \end{array} \right.$$



$$\left\{ \begin{array}{l} \vec{E}_2|_{z=0^-} = E_0'' e^{-i\omega t} \vec{e}_x \\ H_2|_{z=0^-} = -H_0'' e^{-i\omega t} \vec{e}_y = -\frac{1}{Z_2} E_0'' e^{-i\omega t} \vec{e}_y \end{array} \right.$$

垂直入射时:

$$\frac{Z_0'}{Z_0} = \frac{Z_2 - Z_1}{Z_2 + Z_1}, \quad \frac{Z_0''}{Z_0} = \frac{2Z_2}{Z_1 + Z_2}$$

$$\vec{E}_1|_{z=0^+} = \vec{E}_2|_{z=0^-} = \frac{2Z_2}{Z_1 + Z_2} E_0 e^{-i\omega t} \vec{e}_x$$

$$\vec{H}_1|_{z=0^+} = \vec{H}_2|_{z=0^-} = -\frac{2}{Z_1 + Z_2} H_0 e^{-i\omega t} \vec{e}_y$$

$$\begin{aligned} \langle \vec{T} \rangle|_{z=0^+} &= \frac{1}{2} \left[-\epsilon_0 \text{Re}(E_x)^2 + \mu_0 \text{Re}(H_y)^2 \right] (\vec{e}_x \vec{e}_x - \vec{e}_y \vec{e}_y) + \frac{1}{2} \left[\epsilon_0 \text{Re}(E_x)^2 + \mu_0 \text{Re}(H_y)^2 \right] \\ &= T_{xx}|_{z=0^+} (\vec{e}_x \vec{e}_x - \vec{e}_y \vec{e}_y) + T_{zz}|_{z=0^+} \vec{e}_z \vec{e}_z \end{aligned}$$

$$\begin{aligned} \langle \vec{T} \rangle|_{z=0^-} &= \frac{1}{2} \left[-\epsilon \text{Re}(E_x)^2 + \mu_0 \text{Re}(H_y)^2 \right] (\vec{e}_x \vec{e}_x - \vec{e}_y \vec{e}_y) + \frac{1}{2} \left[\epsilon \text{Re}(E_x)^2 + \mu_0 \text{Re}(H_y)^2 \right] \\ &= T_{xx}|_{z=0^-} (\vec{e}_x \vec{e}_x - \vec{e}_y \vec{e}_y) + T_{zz}|_{z=0^-} \vec{e}_z \vec{e}_z \end{aligned}$$

$$\vec{P}_{\text{rad}} = (\langle \vec{T}_{zz} \rangle|_{z=0^-} - \langle \vec{T}_{zz} \rangle|_{z=0^+}) \vec{e}_z$$

$$= \frac{1}{2} (\epsilon - \epsilon_0) \langle \text{Re}(E_x)^2 \rangle \vec{e}_z \quad \langle \text{Re}(E_x)^2 \rangle = \frac{1}{2} \text{Re}(E_x)^2$$

$$= \frac{1}{4} (\epsilon - \epsilon_0) \left(\frac{2Z_2}{Z_1 + Z_2} \right)^2 E_0^2 \vec{e}_z$$

$$= \left(\frac{n-1}{n+1} \right)^2 \epsilon_0 E_0^2 = 2c\mu_0 \frac{n-1}{n+1} \vec{e}_z = \frac{2Z_0}{c} \frac{n-1}{n+1} \vec{e}_z$$

思考题 1: 在两种介质阻抗相同, 平面电磁波垂直入射时不存在反射波.

全反射时的磁导率介质分界面处 n_2 与分界面垂直的平面。

$$z_1 = z_2 = \sqrt{\frac{\mu_1}{\epsilon_1}} = \sqrt{\frac{\mu_2}{\epsilon_2}}$$

$$\vec{r}_{\text{prod}} = \frac{2j_0}{c} \frac{n-1}{n+1} \vec{e}_z = \frac{2j_0}{c} \frac{(\sqrt{\epsilon_1 \mu_1} - 1)}{(\sqrt{\epsilon_2 \mu_2} - 1)} \vec{e}_z \neq 0$$

6. Brewster角 $(\frac{\vec{E}_0'}{\vec{E}_0})_s = \frac{z_2 \cos \theta - z_1 \cos \theta''}{z_2 \cos \theta + z_1 \cos \theta''}$, $(\frac{\vec{E}_0'}{\vec{E}_0})_s = \frac{z_1 \cos \theta - z_2 \cos \theta''}{z_1 \cos \theta + z_2 \cos \theta''}$

$$(\frac{\vec{E}_0'}{\vec{E}_0})_s = 0 \rightarrow z_2 \cos \theta = z_1 \cos \theta''$$

$$n_1 \sin \theta = n_2 \sin \theta''$$

$$\begin{aligned} \hookrightarrow (\frac{z_2}{z_1})^2 \cos^2 \theta + (\frac{n_1}{n_2})^2 \sin^2 \theta &= 1 \\ \tan^2 \theta &= \frac{(z_2/z_1)^2 - 1}{1 - (n_1/n_2)^2} = \frac{\frac{\mu_2 \epsilon_1}{\mu_1 \epsilon_2} - 1}{1 - \frac{\mu_1 \epsilon_1}{\mu_2 \epsilon_2}} = \frac{\mu_2^2 \epsilon_1 - \mu_1 \mu_2 \epsilon_2}{\mu_1 \mu_2 \epsilon_2 - \mu_1^2 \epsilon_1} = \frac{\mu_2}{\mu_1} \frac{\mu_2 \epsilon_1 - \mu_1 \epsilon_2}{\mu_2 \epsilon_2 - \mu_1 \epsilon_1} = -1 \end{aligned}$$

(非磁性介质)

若 $\epsilon_1 = \epsilon_2$, 则垂直入射 / $\theta = 0$ 时, 反射 S 偏振为 0;

对非磁性介质, 不存在 θ 使 S 偏振为 0

$$(\frac{\vec{E}_0'}{\vec{E}_0})_p = 0 \rightarrow z_1 \cos \theta_B = z_2 \cos \theta''$$

$$\begin{aligned} \hookrightarrow (\frac{z_1}{z_2})^2 \cos^2 \theta_B + (\frac{n_1}{n_2})^2 \sin^2 \theta_B &= 1 \\ \tan^2 \theta_B &= \frac{(z_1/z_2)^2 - 1}{1 - (n_1/n_2)^2} = \frac{\frac{\mu_1 \epsilon_2}{\mu_2 \epsilon_1} - 1}{1 - \frac{\mu_1 \epsilon_1}{\mu_2 \epsilon_2}} = \frac{\mu_1 \epsilon_2^2 - \mu_2 \epsilon_1 \epsilon_2}{\mu_2 \epsilon_1 \epsilon_2 - \mu_1 \epsilon_1^2} = \frac{\epsilon_2}{\epsilon_1} \frac{\mu_1 \epsilon_2 - \mu_2 \epsilon_1}{\mu_2 \epsilon_2 - \mu_1 \epsilon_1} = \frac{\epsilon_2}{\epsilon_1} \end{aligned}$$

(非磁性介质)

$$\Rightarrow \theta_B = \arctan \frac{n_2}{n_1}, \quad \theta_B + \theta'' = 90^\circ$$

7. 全反射现象.

$$\vec{E}_1' \rightarrow -\vec{E}_1'$$

$$(\frac{\vec{E}_0'}{\vec{E}_0})_p = -\frac{\sqrt{\epsilon_2} \cos \theta - \sqrt{\epsilon_1} \cos \theta''}{\sqrt{\epsilon_2} \cos \theta + \sqrt{\epsilon_1} \cos \theta''}$$

临界角 $\theta = \theta_c$, $\theta'' = 90^\circ$ 解得 $\sin \theta_c = \frac{n_2}{n_1} = \sqrt{\frac{\epsilon_2}{\epsilon_1}}$

$\theta > \theta_c$ 时:

$$(\frac{\vec{E}_0'}{\vec{E}_0})_{p/s} \rightarrow \text{虚数}$$

$$\cos \theta'' = i \sqrt{\sin^2 \theta - 1} = i \sqrt{\frac{\sin^2 \theta}{\sin^2 \theta_c} - 1}$$

$$(\frac{\vec{E}_0'}{\vec{E}_0})_s = \frac{\cos \theta - i \sqrt{\frac{\epsilon_2}{\epsilon_1} \left(\frac{\sin^2 \theta}{\sin^2 \theta_c} - 1 \right)}}{\cos \theta + i \sqrt{\frac{\epsilon_2}{\epsilon_1} \left(\frac{\sin^2 \theta}{\sin^2 \theta_c} - 1 \right)}} = \frac{\cos \theta - i \sqrt{\sin^2 \theta - \sin^2 \theta_c}}{\cos \theta + i \sqrt{\sin^2 \theta - \sin^2 \theta_c}}$$

$$z' = \frac{\cos \theta - i \sqrt{\sin^2 \theta - \sin^2 \theta_c}}{\cos \theta + i \sqrt{\sin^2 \theta - \sin^2 \theta_c}}$$

$$\left(\frac{\vec{E}_0}{\vec{E}_0}\right)_p = -\cos\theta + i \sqrt{\frac{\epsilon_1}{\epsilon_2}} \left(\frac{\sin^2\theta}{\sin^2\theta_c} - 1 \right) = -\cos\theta \sin^2\theta_c + i \sqrt{\sin^2\theta - \sin^2\theta_c}$$

$$\theta > \theta_c, \quad \left|\frac{\vec{E}_0'}{\vec{E}_0}\right|_s = 1, \quad \left(\frac{\vec{E}_0'}{\vec{E}_0}\right)_p = 1$$

8. 全反射下的消逝波

$$\text{若 } n_1 > n_2, \quad k_x'' = k_1 \frac{n_2}{n_1} = k_1 \sin\theta_c, \quad k_z'' = n_2 k_0, \quad k_1 = n_1 k_0$$

$$\text{边界条件: } k_x'' = k_x = k_1 \sin\theta > k_x'' \quad (\text{全反射, } \theta > \theta_c)$$

$$\text{此时 } k_z'' = \sqrt{k_x''^2 - k_x'^2} = i k_2 \sqrt{\sin^2\theta - \sin^2\theta_c}$$

$$= i k_2 \quad k = k_2 \sqrt{\sin^2\theta - \sin^2\theta_c}$$

$$\vec{E}''(\vec{r}, t) = \vec{E}_0'' e^{-kz} e^{i(k_x x - \omega t)}$$

$$\text{相速度 } v_p = \frac{\omega}{k_x''} = \frac{\omega}{k_x} = \frac{\omega}{k_1 \sin\theta} = \frac{c}{n \sin\theta} \rightarrow v_p = \frac{c}{n}$$

$$v_{up} = \frac{c}{n_2} > \frac{c}{n} \cdot \frac{\sin\theta_c}{\sin\theta} = \frac{c}{n \sin\theta} = v_p$$

$$v_{up} > v_p > v_{ip}$$

$$\text{解法: } \vec{E}''(\vec{r}, t) = E_0'' e^{-kz} e^{i(k_x x - \omega t)} \vec{e}_y = \vec{E}_y'' \vec{e}_y$$

$$\vec{B} = \frac{1}{\omega} \vec{r} \times \vec{E} \quad k_x'' = k_x = k_1 \sin\theta = \frac{k_1 n_2}{n_1} \sin\theta = k_1 \frac{\sin\theta}{\sin\theta_c}$$

$$B_z'' = \frac{k_x''}{\omega} E_y'' = \frac{k_x'}{\omega} \frac{\sin\theta}{\sin\theta_c} E_y'' = \sqrt{\epsilon_1 \mu_2} \frac{\sin\theta}{\sin\theta_c} E_y''$$

$$B_z'' = -\frac{k_z''}{\omega} E_y'' = -\frac{i k_2}{\omega} E_y''$$

B_z'' 与 E_y'' 同相位, B_x'' 与 E_y'' 相位差。

$$\text{Re}(B_z'') = \sqrt{\mu_2 \epsilon_1} \frac{\sin\theta}{\sin\theta_c} \text{Re}(E_y''), \quad \text{Re}(B_x'') = \frac{k_z}{\omega} \text{Im}(E_y'')$$

$$\vec{S} = \vec{E} \times \vec{H}$$

$$S_{yz} = -\text{Re}(E_y'') \frac{\text{Re}(B_x'')}{\mu_2}$$

$$= -\frac{k_z}{\mu_2 \omega} \text{Re}(E_y'') \text{Im}(E_y'')$$

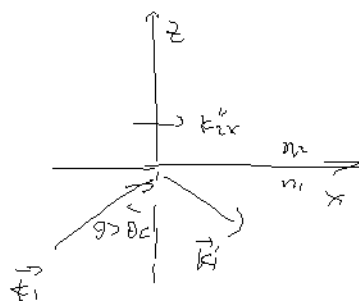
$$= -\frac{k_z}{\mu_2 \omega} |\vec{E}_0''|^2 e^{-2kz} \cos(k_z z - \omega t) \sin(k_z z - \omega t)$$

$$= -\frac{k_z}{2\mu_2 \omega} |\vec{E}_0''|^2 e^{-2kz} \sin[2(k_z z - \omega t)]$$

$\langle S_{yz} \rangle = 0$ 消逝波沿 z 方向平均射流为 0

$$S_{yx} = \text{Re}(E_y'') \text{Re}(B_z'') \cdot \frac{1}{\mu_2}$$

$$= \sqrt{\epsilon_1} \frac{\sin\theta}{\mu_2 \sin\theta_c} |\vec{E}_0''|^2 e^{-2kz} \cos^2(k_z z - \omega t)$$



$$\langle S_{zx} \rangle = \frac{1}{2} \sqrt{\frac{\epsilon_1}{\mu_2}} \frac{\sin \theta}{\sin \theta_c} |\vec{E}_0|^2 e^{-2kz}$$

9. 负折射率和负折射现象

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \quad \frac{1}{\mu} \nabla \times \vec{B} = \epsilon \frac{\partial \vec{E}}{\partial t}$$

$$\text{单色平面波: } \vec{E} \times \vec{E} = \omega \vec{B}, \quad \frac{1}{\mu} \vec{E} \times \vec{B} = -\omega \epsilon \vec{E}$$

$$\hookrightarrow \frac{n}{c} \vec{E} \times \vec{E} = \vec{B}, \quad \frac{n}{c\mu} \vec{E} \times \vec{B} = -\epsilon \vec{E}$$

$$\text{其中 } \vec{E} = n \vec{E}_0 = \frac{n\omega}{c} \vec{E}_k, \quad n = \sqrt{\epsilon_r \mu_r}$$

双负材料: $\epsilon_r < 0, \mu_r < 0$

$$n = -\sqrt{\epsilon_r \mu_r}$$

$(\vec{E}_k, \vec{E}, \vec{B})$ 构成左手关系

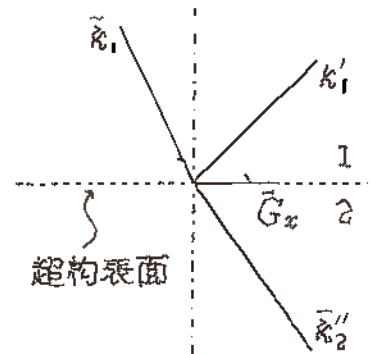
10. 超构表面

假设表面产生相位延迟 $G(x)$, 相位延迟 $G(x) = \frac{d\Phi(x)}{dx}$

$$\begin{cases} k_{1x}' = k_{1x} + G_x & k_{1x} = n_1 k_0 \sin \theta, \quad k_{1x}' = n_1 k_0 \sin \theta' \\ k_{2x}'' = k_{1x} + G_x & k_{2x}'' = n_2 k_0 \sin \theta'' \end{cases}$$

$$\begin{cases} n_1 \sin \theta' = n_1 \sin \theta + \frac{G_x}{k_0} \\ n_2 \sin \theta'' = n_1 \sin \theta + \frac{G_x}{k_0} \end{cases}$$

产生新奇折射、反射现象。



§ 5.3 电磁波在多层介质膜中的传播.

1. 单层介质增透膜

$$\vec{E}_z = E_{z0} e^{i(k_1 z - \omega t)} \vec{e}_x, \quad \vec{E}_R = E_{R0} e^{i(-k_1 z - \omega t)} \vec{e}_x$$

$$\vec{E}_+ = E_{+0} e^{i(k_2(z-d) - \omega t)} \vec{e}_x, \quad \vec{E}_- = E_{-0} e^{i(-k_2(z-d) - \omega t)} \vec{e}_x$$

$$\vec{E}_T = E_{T0} e^{i(k_3(z-d) - \omega t)} \vec{e}_x$$

边界条件:

$$z=0: \begin{cases} E_{z0} + E_{R0} = E_{+0} e^{-ik_2 d} + E_{-0} e^{ik_2 d} \\ \frac{1}{Z_1} (E_{z0} - E_{R0}) = \frac{1}{Z_2} (E_{+0} e^{-ik_2 d} - E_{-0} e^{ik_2 d}) \end{cases}$$

$$\phi_2 = k_2 d = n_2 d k_0 = n_2 d \frac{\omega}{c}$$

$$\begin{pmatrix} E_{z0} \\ E_{R0} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} (1 + \frac{Z_1}{Z_2}) e^{-i\phi_2} & \frac{1}{2} (1 - \frac{Z_1}{Z_2}) e^{i\phi_2} \\ \frac{1}{2} (1 - \frac{Z_1}{Z_2}) e^{-i\phi_2} & \frac{1}{2} (1 + \frac{Z_1}{Z_2}) e^{i\phi_2} \end{pmatrix} \begin{pmatrix} E_{+0} \\ E_{-0} \end{pmatrix}$$

$$z=d: \begin{cases} E_{+0} + E_{-0} = E_{T0} \\ \frac{1}{Z_2} (E_{+0} - E_{-0}) = \frac{1}{Z_3} E_{T0} \end{cases}$$

$$\text{解得} \begin{cases} E_{+0} = \frac{1}{2} (1 + \frac{Z_2}{Z_3}) E_{T0} \\ E_{-0} = \frac{1}{2} (1 - \frac{Z_2}{Z_3}) E_{T0} \end{cases}$$

$$\hookrightarrow E_{+0} (1 - \frac{n_2^2}{n_3^2}) + E_{-0} (1 + \frac{n_2^2}{n_3^2}) = 0$$

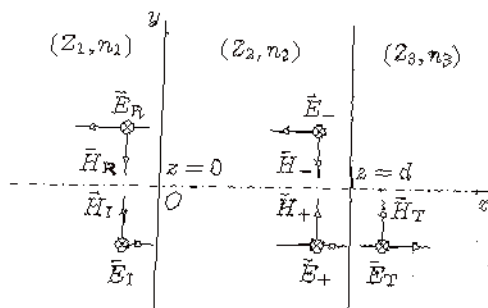
要求增透, $E_{R0} = 0$

$$E_{+0} (1 - \frac{n_2^2}{n_1^2}) e^{-i\phi_2} + E_{-0} (1 + \frac{n_2^2}{n_1^2}) e^{i\phi_2} = 0$$

(E_{+0}, E_{-0}) 的非零解要求:

$$\begin{vmatrix} 1 - \frac{n_2^2}{n_1^2} & 1 + \frac{n_2^2}{n_1^2} \\ (1 - \frac{n_2^2}{n_1^2}) e^{-i\phi_2} & (1 + \frac{n_2^2}{n_1^2}) e^{i\phi_2} \end{vmatrix} = 0$$

$$\hookrightarrow \begin{cases} n_1 \cos \phi_2 = n_3 \cos \phi_2 \end{cases}$$



$$n_2^2 \sin \phi_1 = n_1 n_3 \sin \phi_2$$

$$n_1 \neq n_3: \quad \begin{cases} \phi_2 = k_2 d = \frac{1}{2}(2m+1)\lambda \\ n_2 = \sqrt{n_1 n_3} \end{cases} \quad \begin{cases} n_2 d = \frac{1}{2}(2m+1)\lambda_0 \\ n_2 = \sqrt{n_1 n_3} \end{cases}$$

$$n_1 = n_3: \quad \phi_2 = m\lambda, \quad k_2 = \frac{n_2 \pi}{d}$$

Fabry - Perot 共振腔

2. Fabry - Perot 共振腔

上述 $n_1 = n_3$ 的情况, 同色入射光波长 λ_0

$$\text{膜厚度 } d = \frac{n_2 \lambda_0}{2n_2} \quad / \quad \text{频率 } \nu = \frac{c}{\lambda_0} \rightarrow m \frac{c}{2d}$$

窄带频的透射效应

$$\begin{pmatrix} E_{T0} \\ E_{R0} \end{pmatrix} = \begin{pmatrix} \frac{1}{2}(1 + \frac{n_2}{n_1})e^{-i\phi_1} & \frac{1}{2}(1 - \frac{n_2}{n_1})e^{i\phi_1} \\ \frac{1}{2}(1 - \frac{n_2}{n_1})e^{-i\phi_2} & \frac{1}{2}(1 + \frac{n_2}{n_1})e^{i\phi_2} \end{pmatrix} \begin{pmatrix} E_{I0} \\ E_{R0} \end{pmatrix}$$

$$E_{T0} = \frac{1}{2}(1 + \frac{n_2}{n_1})E_{I0}, \quad E_{R0} = \frac{1}{2}(1 - \frac{n_2}{n_1})E_{I0}$$

$$\text{定义 } R = \left(\frac{n_1 - n_2}{n_1 + n_2} \right)^2$$

介质膜透射率

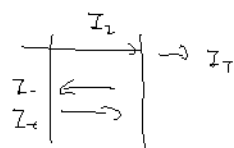
$$t = \left| \frac{E_{T0}}{E_{I0}} \right|^2 = \left| 1 + \frac{4R}{(1-R)^2} \sin^2 \phi_2 \right|^{-1} \quad \phi_1 = k_1 d \quad \phi_2 = m\lambda, \text{ 共振, } t = 1$$

R-P 标准具

$$I_T = |t|^2 I_1, \quad I_R = (1-R) I_1, \quad I_- = R I_+$$

$$\text{求解 } I_+ = \frac{(1-R)}{1-R} I_2, \quad I_- = \frac{|t|^2 R}{1-R} I_2$$

$$\text{透射频率 } \nu_m = m \frac{c}{2d}$$



3. 多层介质膜反射、Brugg反射.

$$(n_1, z_1) \quad (n_2, z_2) \quad (n_3, z_3)$$

$$\begin{cases} E_1 = E_{10} + E_{1R} \\ H_1 = H_{10} + H_{1R} \end{cases} \quad \begin{cases} E_2 = E_{20} \\ H_2 = H_{20} \end{cases}$$

$$\begin{matrix} \rightarrow & \rightarrow \\ (E_1, H_1) & (E_2, H_2) \end{matrix}$$

;

$$\begin{pmatrix} E_1 \\ H_1 \end{pmatrix} = \begin{pmatrix} \cos(k_2 d) & -i z_2 \sin(k_2 d) \\ -\frac{i}{z_2} \sin(k_2 d) & \cos(k_2 d) \end{pmatrix} \begin{pmatrix} E_2 \\ H_2 \end{pmatrix}$$

$$\begin{pmatrix} E_1 \\ H_1 \end{pmatrix} = \prod_{j=1}^{N-1} M_j \begin{pmatrix} E_N \\ H_N \end{pmatrix} \quad \text{其中 } M_j = \begin{pmatrix} \cos \phi_j & -i z_j \sin \phi_j \\ -\frac{i}{z_j} \sin \phi_j & \cos \phi_j \end{pmatrix} \quad \phi_j = k_j d_j = n_j d_j k_0$$

$$\text{定义 } r = \frac{E_{1R}}{E_{10}}, \quad t = \frac{E_{20}}{E_{10}}, \quad E_{20} = 1, \quad H_{10} = \frac{1}{z_1}$$

$$\text{则 } \begin{cases} E_1 = 1+r, & H_1 = \frac{1+r}{z_1} \\ E_N = t, & H_N = \frac{t}{z_N} \end{cases}$$

$$\begin{pmatrix} 1+r \\ \frac{1+r}{z_1} \end{pmatrix} = \prod_{j=1}^{N-1} M_j \begin{pmatrix} t \\ \frac{t}{z_N} \end{pmatrix}$$

考虑周期性结构, 周期数为L

$$\begin{pmatrix} E_1 \\ H_1 \end{pmatrix} = \begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix}^L \begin{pmatrix} E_{L+1} \\ H_{L+1} \end{pmatrix}$$

$$m_{11} = \cos \phi_a \cos \phi_b - \frac{z_a}{z_b} \sin \phi_a \sin \phi_b$$

$$m_{12} = -i (z_b \cos \phi_a \sin \phi_b + z_a \sin \phi_a \cos \phi_b)$$

$$m_{21} = -i (\frac{1}{z_a} \sin \phi_a \cos \phi_b + \frac{1}{z_b} \cos \phi_a \sin \phi_b)$$

$$m_{22} = \cos \phi_a \cos \phi_b - \frac{z_b}{z_a} \sin \phi_a \sin \phi_b$$

$$\phi_a = k_a d_a = n_a d_a k_0, \quad \phi_b = k_b d_b = n_b d_b k_0$$

若使 $n_a d_a = n_b d_b = n d$, $\phi = \phi_a = \phi_b$

$$\begin{pmatrix} E_1 \\ H_1 \end{pmatrix} = \begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix}^L \begin{pmatrix} E_{L+1} \\ H_{L+1} \end{pmatrix}$$

$$\begin{pmatrix} E_1 \\ H_1 \end{pmatrix} = \begin{pmatrix} \frac{z_b}{z_a} & 0 \\ 0 & -\frac{z_b}{z_a} \end{pmatrix} \begin{pmatrix} E_N \\ H_N \end{pmatrix}$$

$$\text{若 } k_0 \text{ 满足 } k_0 \cdot n d = \phi = \frac{2m\pi}{2} \Rightarrow$$

$$\begin{pmatrix} 1+r \\ \frac{1-r}{z_1} \end{pmatrix} = \begin{pmatrix} -\frac{z_b}{z_a} & 0 \\ 0 & -\frac{z_b}{z_a} \end{pmatrix} \begin{pmatrix} 1 \\ \frac{1}{z_N} \end{pmatrix}$$

$$\text{若 } n_1 = n_N, z_1 = z_N$$

$$\begin{cases} 1+r = (-\frac{z_b}{z_a})^L \cdot 1 \\ 1-r = (-\frac{z_b}{z_a})^L \cdot 1 \end{cases}$$

$$\text{对非磁性介质, } \mu_c = \mu_b = \mu_0, \frac{z_a}{z_b} = \sqrt{\frac{\mu_0 \epsilon_a}{\mu_0 \epsilon_b}} = \frac{n_b}{n_a}$$

$$\text{解得 } r = \frac{(\frac{n_b}{n_a})^L - (\frac{n_b}{n_a})^{-L}}{(\frac{n_b}{n_a})^L + (\frac{n_b}{n_a})^{-L}}$$

$$L \gg 1 \text{ 时 } |r| \rightarrow 1$$

$$\text{讨论: } k_a d_a = k_b d_b = \frac{\pi}{2}$$

$$\text{此时 } n_a d_a = n_b d_b = \frac{\pi}{2}, \frac{1}{\lambda_0} = \frac{1}{4} \lambda_{\text{gap}, m=0}$$

$$\text{高反频率 } \frac{1}{\lambda_{\text{gap}}} = \frac{2m+1}{\lambda_{\text{gap}, m=0}}$$

4. 一维光子晶体相邻周期单元的电磁场

折射率呈一个方向周期性变化。

$$\text{设晶体周期为 } L = d_a + d_b$$

$$\text{周期性: } \epsilon(z) = \epsilon(z+L), \mu(z) = \mu(z+L)$$

$$\text{设电场为 } \vec{E}(y, z, t) = \vec{E}(z) e^{i(k_y y - \omega t)}$$

$$\begin{aligned} E(z) &= E_{L,a}^+ e^{i k_{a2} [z - (L - d_b)]} + E_{L,a}^- e^{-i k_{a2} [z - (L - d_b)]} \quad (L-1)L < z < (L-1)L + d_b \\ &= E_{L,b}^+ e^{i k_{b2} (z - L)} + E_{L,b}^- e^{-i k_{b2} (z - L)} \quad L < z < L + d_b \end{aligned}$$

$$\text{其中 } k_{a2} = \sqrt{k_a^2 - k_y^2} = \sqrt{(\frac{n_a \omega}{c})^2 - k_y^2}, k_{b2} = \sqrt{(\frac{n_b \omega}{c})^2 - k_y^2}$$

1) 5 偏振。

$$\text{电场的 } x \text{ 方向: } \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$H_y = \frac{1}{i\omega\mu} \frac{\partial E_x}{\partial z}$$

在 $z = (l-1)d_b$ 界面,

$$\begin{cases} E_{l,a}^+ e^{ik_{az} d_b} + E_{l,a}^- e^{-ik_{az} d_b} = E_{l-1,b}^+ + E_{l-1,b}^- \\ \frac{k_{az}}{\mu_a} (E_{l,a}^+ e^{ik_{az} d_b} - E_{l,a}^- e^{-ik_{az} d_b}) = \frac{k_{bz}}{\mu_b} (E_{l-1,b}^+ + E_{l-1,b}^-) \end{cases}$$

在 $z = lN - d_b$

$$\begin{cases} E_{l,a}^+ + E_{l,a}^- = E_{l,b}^+ e^{-ik_{bz} d_b} + E_{l,b}^- e^{ik_{bz} d_b} \\ \frac{k_{az}}{\mu_a} (E_{l,a}^+ - E_{l,a}^-) = \frac{k_{bz}}{\mu_b} (E_{l,b}^+ e^{-ik_{bz} d_b} - E_{l,b}^- e^{ik_{bz} d_b}) \end{cases}$$

....

解得
$$\begin{pmatrix} E_{l-1,b}^+ \\ E_{l-1,b}^- \end{pmatrix} = \begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix} \begin{pmatrix} E_{l,b}^+ \\ E_{l,b}^- \end{pmatrix}$$

其中 $m_{11} = e^{-ik_{bz} d_b} \left[\cos(k_{az} d_a) - \frac{i}{2} \left(\frac{k_{az}}{k_{bz}} \frac{\mu_b}{\mu_a} + \frac{k_{bz}}{k_{az}} \frac{\mu_a}{\mu_b} \right) \sin(k_{az} d_a) \right]$

$m_{12} = e^{ik_{bz} d_b} \frac{i}{2} \left(\frac{k_{bz}}{k_{az}} \frac{\epsilon_a}{\epsilon_b} - \frac{k_{az}}{k_{bz}} \frac{\mu_b}{\mu_a} \right) \sin(k_{az} d_a)$

$m_{21} = m_{12}^*$

$m_{22} = m_{11}^*$

$\begin{vmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{vmatrix} = |m_{11}|^2 - |m_{12}|^2 = 1$ 对应电磁波在光子晶体中传播时能量守恒.

12.7 偏振.

磁场沿 y 方向 $\nabla \times \vec{H} = \frac{\partial D}{\partial t}$

$E_x = \frac{i}{\omega \epsilon} \frac{\partial H_x}{\partial z}$

$\begin{pmatrix} H_{l-1,b}^+ \\ H_{l-1,b}^- \end{pmatrix} = \begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix} \begin{pmatrix} H_{l,b}^+ \\ H_{l,b}^- \end{pmatrix}$

其中 $m_{11} = e^{-ik_{bz} d_b} \left[\cos(k_{az} d_a) - \frac{i}{2} \left(\frac{k_{az}}{k_{bz}} \frac{\epsilon_b}{\epsilon_a} + \frac{k_{bz}}{k_{az}} \frac{\epsilon_a}{\epsilon_b} \right) \sin(k_{az} d_a) \right]$

$m_{12} = e^{ik_{bz} d_b} \frac{i}{2} \left(\frac{k_{bz}}{k_{az}} \frac{\epsilon_b}{\epsilon_a} - \frac{k_{az}}{k_{bz}} \frac{\epsilon_b}{\epsilon_a} \right) \sin(k_{az} d_a)$

$m_{21} = m_{12}^*$

$m_{22} = m_{11}^*$

5. 一维光子晶体中的 Bloch 波及光子带隙.

Floquet 定理: 在任意周期性结构中传播的波, 振幅为周期调制的单色平面波形式.

$\vec{E}(y, z, t) = \vec{u}_k(z) e^{ikz} e^{i(k_y y - \omega t)}$

周期性: $\vec{u}_k(z) = \vec{u}_k(z + \Lambda)$

$$\hookrightarrow \begin{pmatrix} E_{1,b}^+ \\ E_{1,b}^- \end{pmatrix} = e^{-ik\Lambda} \begin{pmatrix} E_{1,b}^+ \\ E_{1,b}^- \end{pmatrix} \Rightarrow M \begin{pmatrix} E_{1,b}^+ \\ E_{1,b}^- \end{pmatrix} = e^{-ik\Lambda} \begin{pmatrix} E_{1,b}^+ \\ E_{1,b}^- \end{pmatrix}$$

非零解要求: $\begin{vmatrix} M_{11} - e^{-ik\Lambda} & M_{12} \\ M_{21} & M_{22} - e^{-ik\Lambda} \end{vmatrix} = 0$

解得 $e^{-ik\Lambda} = \frac{1}{2}(M_{11} + M_{22}) \pm \sqrt{\frac{1}{4}(M_{11} + M_{22})^2 - 1}$

$$K(\omega, k_y) = \frac{1}{\Lambda} \arccos \left[\frac{1}{2}(M_{11} + M_{22}) \right]$$

通带: $\frac{1}{2}|M_{11} + M_{22}| < 1$

禁带/带隙: $\frac{1}{2}|M_{11} + M_{22}| > 1$

$$k_{\text{edge}} = \frac{m\pi}{\Lambda} \quad (\frac{1}{2}|M_{11} + M_{22}| \geq 1)$$

S 偏振:

$$\cos(k\Lambda) = \cos(k_{a2} d_a) \cos(k_{b2} d_b) - \frac{1}{2} \left(\frac{k_{a2}}{k_{b2}} \frac{\mu_b}{\mu_a} + \frac{k_{b2}}{k_{a2}} \frac{\mu_a}{\mu_b} \right) \sin(k_{a2} d_a) \sin(k_{b2} d_b)$$

P 偏振:

$$\cos(k\Lambda) = \cos(k_{a2} d_a) \cos(k_{b2} d_b) - \frac{1}{2} \left(\frac{k_{a2}}{k_{b2}} \frac{\epsilon_b}{\epsilon_a} + \frac{k_{b2}}{k_{a2}} \frac{\epsilon_a}{\epsilon_b} \right) \sin(k_{a2} d_a) \sin(k_{b2} d_b)$$

Brewster 现象:

$$\text{当 } k_t = k_0 \left(\frac{1}{n_1^2} + \frac{1}{n_2^2} \right)^{1/2}$$

P 偏振光在通带 $\frac{k_y}{2\pi/\Lambda} = \sqrt{\frac{1}{n_1^2} + \frac{1}{n_2^2}}$

6. 一维光子晶体 Bloch 波的色散特性.

电磁波沿 z 轴传播.

$$k_y = 0, \quad k_{a2} = k_a = \frac{n_a \omega}{c}, \quad k_{b2} = k_b = \frac{n_b \omega}{c}$$

色散关系: $\cos(k\Lambda) = \cos(k_a d_a) \cos(k_b d_b) - \frac{1}{2} \left(\frac{n_a}{n_b} + \frac{n_b}{n_a} \right) \sin(k_a d_a) \sin(k_b d_b)$

§ 0.4 介质及金属导体的色散

1. Lorentz 模型 (介质)

忽略原子间作用力, 带阻尼的谐振子模型

$$m \frac{d^2 \vec{x}}{dt^2} = -m\omega_0^2 \vec{x} - m\gamma \frac{d\vec{x}}{dt} - e\vec{E}_0 e^{-i\omega t}$$

$$\text{解得 } \vec{x} = -\frac{e}{m} \frac{1}{\omega_0^2 - \omega^2 - i\gamma\omega} \vec{E}_0 e^{-i\omega t}$$

$$\text{极化强度: } \vec{P} = -Ne\vec{x} = \frac{Ne^2}{m} \frac{1}{\omega_0^2 - \omega^2 - i\gamma\omega} \vec{E}_0 e^{-i\omega t}$$

$$\chi = \frac{\vec{P}}{\epsilon_0 \vec{E}} = \frac{Ne^2}{\epsilon_0 m} \frac{1}{\omega_0^2 - \omega^2 - i\gamma\omega}$$

$$\epsilon_r = 1 + \chi$$

$$= 1 + \frac{\omega_p^2}{\omega_0^2 - \omega^2 - i\gamma\omega} \quad \omega_p = \sqrt{\frac{Ne^2}{\epsilon_0 m}} \text{ 为等离子体频率}$$

$$= 1 + \frac{\omega_p^2(\omega_0^2 - \omega^2)}{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2} + \frac{i}{2} \frac{\omega_p^2 \gamma \omega}{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2}$$

$$= \epsilon_r' + i\epsilon_r''$$

共振波长: $\omega = \omega_0$, ϵ_r' 增大, 明显吸收.

高频: $\omega \gg \omega_0$, $\epsilon \rightarrow 1$ X射线.

2. Drude 模型 (金属)

$$m \frac{d^2 \vec{x}}{dt^2} = -m\gamma \frac{d\vec{x}}{dt} + (-e\vec{E}_0 e^{-i\omega t})$$

$$\text{解得 } \vec{x} = \frac{e}{m} \frac{1}{\omega^2 + i\gamma\omega} \vec{E}_0 e^{-i\omega t}$$

$$\text{电流密度: } \vec{J} = \rho \vec{v} = -ne \frac{d\vec{x}}{dt}$$

$$= -\frac{ne^2}{m} \frac{1}{\omega^2 + i\gamma\omega} (-i\omega) \vec{E}_0 e^{-i\omega t}$$

$$= \frac{i\omega Ne^2}{m} \frac{1}{\omega^2 + i\gamma\omega} E_0 e^{-i\omega t}$$

$$\text{电子率 } \sigma = \frac{\vec{J}}{E} = \frac{i\omega Ne^2}{m} \frac{1}{\omega^2 + i\gamma\omega}$$

$$= \frac{Ne^2}{m} \frac{1}{\gamma - i\omega} = \frac{\epsilon_0 \omega_p^2}{\gamma - i\omega}$$

直流导电率 $\sigma_0 = \frac{Ne^2}{\gamma m} \quad (\omega \rightarrow 0)$

等离子频率 $\omega_p = \sqrt{\frac{Ne^2}{m\epsilon_0}} = \sqrt{\frac{\sigma_0 \gamma}{\epsilon_0}}$

等效介电常数 $\epsilon_{eff} = \epsilon_0 + \frac{i\sigma}{\omega}$ 金属 $\epsilon_0 = \epsilon_0$

$$\epsilon_{eff} = \epsilon_0 + \frac{i\sigma}{\omega} = \left[\epsilon_0 - \frac{i}{\omega} \text{Im}(\sigma) \right] + \frac{i}{\omega} \text{Re}(\sigma)$$

$$= \epsilon_0 + \frac{iNe^2}{m} \frac{1}{(\gamma\omega - i\omega^2)} = \epsilon_0 - \frac{Ne^2}{m} \frac{1}{\omega^2 + i\gamma\omega}$$

$$\epsilon_r(\omega) = \frac{\epsilon_{eff}}{\epsilon_0} = 1 - \frac{\omega_p^2}{\omega^2 + i\gamma\omega}$$

$$= \epsilon_r' + i\epsilon_r''$$

$$= 1 - \frac{\omega_p^2}{\omega^2 + \gamma^2} + i \frac{\omega_p^2 \gamma}{\omega(\omega^2 + \gamma^2)}$$

1. 低频 $\omega < \gamma, \omega_p$ 无感电, 电感, 远红外

$$\epsilon_r(\omega) = 1 - \frac{\omega_p^2}{\gamma^2} + \frac{i\omega_p^2}{\omega\gamma}$$

$$= 1 - \frac{\epsilon_c}{\epsilon_0\gamma} + i \frac{\epsilon_c}{\epsilon_0\omega}$$

特点:

$$\text{Re}(\epsilon_r) = 1 - \frac{\epsilon_c}{\epsilon_0\gamma} \text{ 与频率无关}$$

$$\text{Im}(\epsilon_r) = \frac{\epsilon_c}{\epsilon_0\omega} \gg 1 \gg \text{Re}(\epsilon_r)$$

$\Rightarrow$ 穿透深度 ≈ 0 , 理想导体.

2. 远红外 / 可见光 $\omega < \omega_p$

$$\epsilon_r(\omega) = \epsilon_r'(\omega) \approx 1 - \frac{\omega_p^2}{\omega^2} < 0 \quad \underline{\text{有}} \text{一定穿透深度.}$$

3. 深紫外, X射线 $\omega \gg \gamma, \omega \gg \omega_p$

$$\epsilon_r(\omega) = \epsilon_r'(\omega)$$

§ 5.5 电磁波在色散媒质中的传播.

1. 色散关系的电磁场结构关系

$$\vec{D}(\vec{R}, t) = \int_{-\infty}^{+\infty} dt' \epsilon(t-t') \vec{E}(\vec{R}, t') \quad \epsilon(t-t') = 0 \quad (t' < t)$$

$$\begin{cases} \vec{E}(\vec{R}, t) = \int_{-\infty}^{+\infty} \vec{E}(\vec{R}, \omega) e^{-i\omega t} d\omega \\ \vec{E}(\vec{R}, t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \vec{E}(\vec{R}, \omega) e^{-i\omega t} d\omega \end{cases}$$

单色电磁波: $\vec{D}(\vec{R}, t) = \vec{D}(\vec{R}) e^{-i\omega t}$, $\vec{E}(\vec{R}, t) = \vec{E}(\vec{R}) e^{-i\omega t}$

$$\vec{D}(\vec{R}) = \epsilon(\omega) \vec{E}(\vec{R}) \int_{-\infty}^{+\infty} dt' \epsilon(t-t') e^{-i\omega(t-t')}$$

线性响应函数 = $\epsilon(\omega) \vec{E}(\vec{R})$

$$\begin{cases} \epsilon(\omega) = \int_{-\infty}^{+\infty} \epsilon(t) e^{-i\omega t} dt \\ \epsilon(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \epsilon(\omega) e^{-i\omega t} d\omega \end{cases}$$

$$\vec{B}(\vec{R}) = \mu(\omega) \vec{H}(\vec{R}), \quad \vec{J}_F(\vec{R}) = \sigma(\omega) \vec{E}(\vec{R})$$

2. 单色平面电磁波在色散媒质中的传播.

$$\begin{cases} \nabla \cdot \vec{E} = 0 \\ \nabla \times \vec{B} = \mu_0 \epsilon \frac{\partial \vec{E}}{\partial t} \\ \nabla \cdot \vec{B} = 0 \\ \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \end{cases} \quad \mu = \mu_0$$

$$\vec{E}(\vec{R}, t) = \vec{E}_0(\vec{R}) e^{i(\vec{R} \cdot \vec{R} - \omega t)}, \quad \vec{B}(\vec{R}, t) = \vec{B}_0(\vec{R}) e^{i(\vec{R} \cdot \vec{R} - \omega t)}$$

$$\begin{cases} \vec{R} \cdot \vec{E} = 0 \\ \vec{R} \times \vec{B}_0 = -\omega \mu_0 \epsilon \vec{E}_0 \\ \vec{R} \cdot \vec{B}_0 = 0 \end{cases}$$

$$\vec{R} \times \vec{E}_0 = \omega \vec{E}_0$$

$$\Rightarrow \vec{R} \times (\vec{R} \times \vec{E}_0) = -\omega^2 \mu_0 \epsilon \vec{E}_0 \quad \text{解得 } k = \sqrt{\mu_0 \epsilon} \omega = k_0 \sqrt{\frac{\epsilon(\omega)}{\epsilon_0}}$$

3. 导电媒质中自由电荷分布特征.

$$\vec{D}(\vec{R}) = \epsilon_0(\omega) \vec{E}(\vec{R})$$

$$\vec{J}_f(\vec{R}) = \sigma_c \vec{E}(\vec{R})$$

$$\text{衰逝特征时间: } \tau = \frac{\epsilon_0}{\sigma_c}$$

4. 单色平面电磁波在导电媒质中的传播.

$$\begin{cases} \nabla \cdot \vec{E} = 0 \\ \nabla \times \vec{B} = \mu_0 (\vec{J}_f + \epsilon_0 \frac{\partial \vec{E}}{\partial t}) = \mu_0 (\sigma_c \vec{E} + \epsilon_0 \frac{\partial \vec{E}}{\partial t}) \\ \nabla \cdot \vec{B} = 0 \\ \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \end{cases}$$

考虑单色平面波 $\vec{E}(\vec{R}, t) = \vec{E}_0(\vec{R}) e^{i(\vec{R} \cdot \vec{R} - \omega t)}$, $\vec{B}(\vec{R}, t) = \vec{B}_0(\vec{R}) e^{i(\vec{R} \cdot \vec{R} - \omega t)}$

$$\begin{cases} \vec{R} \cdot \vec{E}_0 = 0 \\ \vec{R} \times \vec{B}_0 = -\omega \mu_0 (\epsilon_0 + i \frac{\sigma_c}{\omega}) \vec{E}_0 \\ \vec{R} \cdot \vec{B}_0 = 0 \\ \nabla \times \vec{E}_0 = \omega \vec{B}_0 \end{cases}$$

引入等效介电常数 $\epsilon_{eff}(\omega) = \epsilon_0 + i \frac{\sigma_c}{\omega}$

$$k = \omega \sqrt{\mu_0 \epsilon_{eff}(\omega)} = k_0 \sqrt{\frac{\epsilon_{eff}(\omega)}{\epsilon_0}}$$

一般地, 设 $\epsilon(\omega) = \epsilon_0 \epsilon_r' + i \epsilon_0 \epsilon_r''$

$$\text{对应 } \vec{k}(\omega) = \vec{\beta}(\omega) + i \vec{\alpha}(\omega)$$

$$|\vec{k}|^2 = \vec{k} \cdot \vec{k} = \beta^2 - \alpha^2 + 2i \vec{\alpha} \cdot \vec{\beta} = \omega^2 \mu_0 \epsilon$$

$$\begin{cases} \beta^2 - \alpha^2 = \omega^2 \mu_0 \epsilon_0 \epsilon_r' \\ 2 \vec{\beta} \cdot \vec{\alpha} = \omega^2 \mu_0 \epsilon_0 \epsilon_r'' \end{cases}$$

考虑电磁波垂直入射到介质表面. 则 $\vec{\alpha} \perp \vec{\beta}$

$$\begin{cases} \beta^2 - \alpha^2 = \omega^2 \mu_0 \epsilon_0 \epsilon_r' \\ 2 \vec{\beta} \cdot \vec{\alpha} = \omega^2 \mu_0 \epsilon_0 \epsilon_r'' \end{cases} \quad \text{解得 } \begin{cases} \alpha = \frac{\sqrt{2}}{2} \omega \sqrt{\mu_0 \epsilon_0 \epsilon_r'} \left[\left(1 + \left(\frac{\epsilon_r''}{\epsilon_r'} \right)^2 - 1 \right)^{1/2} \right] \\ \beta = \frac{\sqrt{2}}{2} \omega \sqrt{\mu_0 \epsilon_0 \epsilon_r'} \left[\left(1 + \left(\frac{\epsilon_r''}{\epsilon_r'} \right)^2 + 1 \right)^{1/2} \right] \end{cases}$$

$$\begin{aligned}\vec{E}(\vec{r}, t) &= \vec{E}_0(\vec{r}) e^{i(\vec{k} \cdot \vec{r} - \omega t)} \\ &= \vec{E}_0(\vec{r}) e^{-\alpha \vec{r} + i(\vec{k} \cdot \vec{r} - \omega t)}\end{aligned}$$

衰减项

穿透深度 $\delta = \frac{1}{\alpha}$

5. 导电媒质的穿透深度.

$$\epsilon(\omega) = \epsilon_0 + i \frac{\sigma_0}{\omega}$$

$$\begin{cases} \alpha = k_0 \sqrt{\frac{\epsilon_r'}{2}} \left[\sqrt{1 + \left(\frac{\epsilon_r''}{\epsilon_r'}\right)^2} - 1 \right]^{1/2} \\ \beta = k_0 \sqrt{\frac{\epsilon_r'}{2}} \left[\sqrt{1 + \left(\frac{\epsilon_r''}{\epsilon_r'}\right)^2} + 1 \right]^{1/2} \end{cases}$$

$$\begin{cases} \alpha = k_0 \sqrt{\frac{\epsilon_r'}{2}} \left[\sqrt{1 + \left(\frac{\sigma_0}{\epsilon_0 \omega}\right)^2} - 1 \right]^{1/2} \\ \beta = k_0 \sqrt{\frac{\epsilon_r'}{2}} \left[\sqrt{1 + \left(\frac{\sigma_0}{\epsilon_0 \omega}\right)^2} + 1 \right]^{1/2} \end{cases}$$

位移电流: $\vec{J}_D = \frac{\partial \vec{D}}{\partial t} = -i\omega \epsilon_0 \vec{E}$

$$\frac{\vec{J}_D}{\vec{J}_0} = \left| \frac{\sigma_0 \vec{E}}{-i\omega \epsilon_0 \vec{E}} \right| = \frac{\sigma_0}{\omega \epsilon_0}$$

① 弱导电媒质: 导电率较小 $\frac{\sigma_0}{\omega \epsilon_0} \ll 1$

$$\begin{cases} \alpha \approx \frac{1}{2} k_0 \frac{\sigma_0}{\omega \epsilon_0} \sqrt{\frac{\epsilon_0}{\epsilon_r}} & \alpha \ll \beta \\ \beta \approx k_0 \sqrt{\frac{\epsilon_0}{\epsilon_r}} \end{cases}$$

相速度: $v_g = \frac{\omega}{\beta} = c \sqrt{\frac{\epsilon_0}{\epsilon_r}}$

② 良导电媒质: $\frac{\sigma_0}{\omega \epsilon_0} \gg 1$

$$\alpha \approx \beta \approx k_0 \sqrt{\frac{\sigma_0}{2\omega \epsilon_0}}$$

6. 导电媒质中的流密度与Joule热.

$$\text{折射率 } \vec{k}'' = \vec{\beta} + i\alpha = (\beta + i\alpha) \vec{e}_z$$

$$\vec{E}''(\vec{r}, t) = \vec{E}_0'' e^{i\vec{k}'' \cdot \vec{r} - i\omega t} = E_0'' e^{-\alpha z} e^{i(\beta z - \omega t)} \vec{e}_y$$

$$\vec{H}'' = \frac{1}{i\omega\mu_0} \nabla \times \vec{E} = \frac{\vec{k}''}{\omega\mu_0} \times \vec{E}''$$

$$= -\frac{1}{\omega\mu_0} (\beta + i\alpha) E_0'' \vec{e}_x$$

$$\text{流密度: } \vec{S}''(\vec{r}, t) = \text{Re}(\vec{E}'') \times \text{Re}(\vec{H}'')$$

$$= \frac{1}{\omega\mu_0} e^{-2\alpha z} E_0''^2 \text{Re}[(\beta + i\alpha) e^{i(\beta z - \omega t)}] \cos(\beta z - \omega t) \vec{e}_z$$

$$= \frac{E_0''^2}{\omega\mu_0} \frac{e^{-2\alpha z}}{|\alpha^2 + \beta^2|} \cos\left[\arctan\frac{\alpha}{\beta} + \beta z - \omega t\right] \cos(\beta z - \omega t)$$

$$= \frac{E_0''^2}{\omega\mu_0} \left[\beta \cos(\beta z - \omega t) - \alpha \sin(\beta z - \omega t) \right] \cos(\beta z - \omega t)$$

$$\langle \vec{S}'' \rangle = \frac{\beta}{2\omega\mu_0} E_0''^2 e^{-2\alpha z} \vec{e}_z$$

流以导体的能流最终转化为Joule热.

$$W = \vec{J}_p \cdot \vec{E} = \sigma_c \vec{E}'' \cdot \vec{E}''$$

$$\text{Joule热功率 } \langle W \rangle = \langle \sigma_c \text{Re}(\vec{E}'') \cdot \text{Re}(\vec{E}'') \rangle$$

$$= \frac{1}{2} \sigma_c (E_0'')^2 e^{-2\alpha z}$$

单位表面积Joule热功率.

$$P = \int_0^\infty \langle W \rangle dz = \frac{\sigma_c}{4\alpha} (E_0'')^2$$

$$= \langle S'' \rangle|_{z=0} = \frac{\beta}{2\omega\mu_0} (E_0'')^2 \quad 2\alpha\beta = \omega^2 \epsilon_0 \mu_0 \epsilon_r = \omega^2 \epsilon_0 = \frac{\sigma_c}{c}$$

$$= \omega \mu_0 \epsilon_0$$

进入导体损耗的能量 $\propto$ 在表面一层厚 δ 的电阻损耗的能量.

$$\propto \times \text{表面电阻 } R_s = \frac{L}{\sigma \delta S} = \frac{1}{\sigma \delta} = \frac{\alpha}{\epsilon_0}$$

7. 导电媒质的反射率.

$$\text{边界条件} \begin{cases} E + E' = E'' \\ H - H' = H'' \end{cases}$$

$$\nabla \times \vec{E} = i\omega \mu_0 \vec{H}$$

$$\vec{H} = -\frac{k_0 \epsilon}{\omega \mu_0} \vec{E}_x \quad \vec{H}' = \frac{k_0 \epsilon}{\omega \mu_0} \vec{E}_x$$

$$\vec{H}'' = -\frac{(\beta + i\alpha) \epsilon''}{\omega \mu_0} \vec{E}_x$$

$$\begin{cases} E + E' = E'' \\ E - E' = \frac{1}{k_0} (\beta + i\alpha) E'' \end{cases} \quad \text{求解 } \frac{E'}{E} = \frac{k_0 - \beta - i\alpha}{k_0 + \beta + i\alpha}$$

$$\text{反射系数 } R = \left| \frac{E'}{E} \right|^2 = \frac{(k_0 - \beta)^2 + \alpha^2}{(k_0 + \beta)^2 + \alpha^2}$$

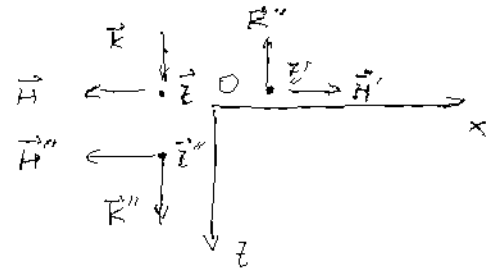
① 弱导电媒质

$$\alpha \ll \beta, \quad R = \left(\frac{k_0 - \beta}{k_0 + \beta} \right)^2 \approx \left(\frac{\sqrt{\epsilon_0} - \sqrt{\epsilon_0}}{\sqrt{\epsilon_0} + \sqrt{\epsilon_0}} \right)^2 \quad \beta \approx k_0 \sqrt{\frac{\epsilon_0}{\epsilon_1}}$$

② 良导电媒质

$$\alpha \approx \beta \gg k_0, \quad R = \frac{(1 - \frac{k_0^2}{\beta^2})^2 + 1}{(1 + \frac{k_0^2}{\beta^2})^2 + 1} \approx \frac{2 - 2\frac{k_0}{\beta}}{2 + 2\frac{k_0}{\beta}} \approx 1 - \frac{2k_0}{\beta} \approx 1 - \sqrt{\frac{8\omega\epsilon_0}{\sigma}}$$

8. 导电薄膜的光吸收率.



9. 金属导体的穿透深度, ($\omega < \omega_p$)

Drude模型:

$$\epsilon(\omega) = \epsilon_0 \epsilon_r' + i\epsilon_0 \epsilon_r''$$

$$\text{其中 } \epsilon_r' = 1 - \frac{\omega_p^2}{\omega^2 + \gamma^2}, \quad \epsilon_r'' = \frac{\omega_p^2 \gamma}{\omega(\omega^2 + \gamma^2)} \quad \gamma \ll \omega \rightarrow \epsilon_r' < 0$$

$$\text{此时可得 } \begin{cases} \alpha = ik_0 \sqrt{\frac{\epsilon_r''}{2}} \left[\sqrt{1 + \left(\frac{\epsilon_r''}{\epsilon_r'}\right)^2 + 1} \right]^{1/2} = -\alpha' \\ \beta = -ik_0 \sqrt{\frac{\epsilon_r''}{2}} \left[\sqrt{1 + \left(\frac{\epsilon_r''}{\epsilon_r'}\right)^2 - 1} \right]^{1/2} = \beta' \end{cases}$$

$$\vec{k}'' = (\alpha' + i\beta') \vec{e}_z$$

$$\vec{E}''(\vec{r}, t) = \vec{E}_0'' e^{i(\vec{k}'' \cdot \vec{r} - \omega t)} = \vec{E}_0'' e^{-\beta' z} e^{i(\alpha' z - \omega t)} \vec{e}_y$$

$$\text{穿透深度 } \delta' = \frac{1}{\beta'}$$

$$\vec{H}''(\vec{r}, t) = \frac{1}{\omega \mu_0} \vec{k} \times \vec{E}'' = -\frac{1}{\omega \mu_0} (\alpha' + i\beta') \vec{E}_0'' e^{-\beta' z} e^{i(\alpha' z - \omega t)} \vec{e}_x$$

平均能流密度:

$$\begin{aligned} \langle \vec{J} \rangle &= \langle \text{Re}(\vec{E}'') \times \text{Re}(\vec{H}'') \rangle \\ &= \frac{\alpha}{\omega \mu_0} |\vec{E}_0''|^2 e^{-2\beta' z} \cos[2(\alpha' z - \omega t)] \vec{e}_z \\ &= \frac{\alpha}{2\omega \mu_0} |\vec{E}_0''|^2 e^{-2\beta' z} \vec{e}_z \end{aligned}$$

$$\text{金属: } \epsilon_r'' = \frac{\text{Re}(G)}{\epsilon_0 \omega}$$

$$\epsilon_r'' \neq 0 \rightarrow \alpha' \neq 0 \rightarrow \langle \vec{J} \rangle \neq 0$$

Re(6)

金属电导率的实部对应损耗.

§ 5.6 波导及谐振腔.

1. 理想导体表面电磁场的特点.

$$E_t = 0, B_n = 0$$

$$\begin{cases} \vec{K}_s = \vec{n} \times \vec{H} \\ \sigma_s = \vec{n} \cdot \vec{D} \end{cases}$$

2. 电磁波导的特点.

考虑在介质 μ, ϵ 中传播的时谐电磁波.

$$\vec{E}(\vec{r}, t) = \vec{E}(\vec{r}) e^{-i\omega t}, \quad \vec{B}(\vec{r}, t) = \vec{B}(\vec{r}) e^{-i\omega t}$$

$$\begin{cases} \nabla^2 \vec{E}(\vec{r}) + K^2 \vec{E}(\vec{r}) = 0 & K = \omega \sqrt{\mu\epsilon} \\ \nabla \cdot \vec{E}(\vec{r}) = 0 \\ \nabla^2 \vec{B}(\vec{r}) + K^2 \vec{B}(\vec{r}) = 0 \\ \nabla \cdot \vec{B}(\vec{r}) = 0 \end{cases}$$

波导轴沿 z 方向:

$$\begin{cases} \vec{E}(\vec{r}, t) = \vec{E}(x, y) e^{i(k_z z - \omega t)} \\ \vec{H}(\vec{r}, t) = \vec{H}(x, y) e^{i(k_z z - \omega t)} \end{cases}$$

波动方程:

$$\nabla^2 \vec{E}(x, y) + K_z^2 \vec{E}(x, y) = 0 \quad K_z^2 = K^2 - k_z^2$$

$$(\partial_x^2 + \partial_y^2 + k^2) \vec{H}(x, y) = 0$$

解得

$$\begin{cases} E_x(x, y) = \frac{i}{k^2} (k_y \frac{\partial^2 \vec{E}}{\partial x^2} + \omega \mu \frac{\partial H_z}{\partial y}) \\ E_y(x, y) = \frac{i}{k^2} (k_x \frac{\partial^2 \vec{E}}{\partial y^2} - \omega \mu \frac{\partial H_z}{\partial x}) \\ H_x(x, y) = \frac{i}{k^2} (-\omega \epsilon \frac{\partial^2 \vec{E}}{\partial y^2} + k_y \frac{\partial H_z}{\partial x}) \\ H_y(x, y) = \frac{i}{k^2} (\omega \epsilon \frac{\partial^2 \vec{E}}{\partial x^2} + k_x \frac{\partial H_z}{\partial y}) \end{cases}$$

假设波导中存在TEM导模

$$E_z = 0, B_z = 0, k_z = k$$

$$\vec{E}(\vec{r}, t) = [E_{0x}(x, y) \vec{e}_x + E_{0y}(x, y) \vec{e}_y] e^{i(k_z z - \omega t)}$$

$$\vec{E}_0(x, y) = E_{0x}(x, y) \vec{e}_x + E_{0y}(x, y) \vec{e}_y$$

$$\begin{aligned} \nabla \times \vec{E}_0(x, y) &= -\frac{\partial E_{0y}}{\partial x} \vec{e}_z + \frac{\partial E_{0x}}{\partial y} \vec{e}_z + \left(\frac{\partial E_{0y}}{\partial x} - \frac{\partial E_{0x}}{\partial y} \right) \vec{e}_z \\ &= \left(\frac{\partial E_{0y}}{\partial x} - \frac{\partial E_{0x}}{\partial y} \right) \vec{e}_z \end{aligned}$$

$$\nabla \times \vec{E}_0(x, y) = (\nabla \times \vec{E})_z = -\frac{\partial B_z}{\partial t} = 0$$

$$\hookrightarrow \vec{E}_0(x, y) = -\nabla \varphi(x, y)$$

$$\nabla^2 \varphi(x, y) = 0 \quad \nabla \cdot \vec{E}_0(x, y) = 0$$

$$\text{边界条件: } E_t = 0$$

→ 单边 (矩形, 圆形 ...), $\varphi = \text{const}$ 不存在 TEM

双边 (平行板), 存在 TEM

3. 矩形波导管中的TE导模 (横电)

$$E_z = 0, H_z \neq 0$$

E_x, E_y, H_x, H_y 只与 H_z 有关.

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) H_z(x, y) + k_c^2 H_z(x, y) = 0$$

分离变量: $H_z(x, y) = X(x) Y(y)$

$$\hookrightarrow \frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} + k_c^2 = 0$$

$$\begin{cases} \frac{d^2 X}{dx^2} + k_x^2 X = 0 \\ \frac{d^2 Y}{dy^2} + k_y^2 Y = 0 \end{cases} \quad k_x^2 + k_y^2 = k_c^2$$

$$\text{解得 } H_z(x, y) = [C_1 \cos(k_x x) + D_1 \sin(k_x x)] [C_2 \cos(k_y y) + D_2 \sin(k_y y)]$$

$$\begin{cases} E_x(x, y) = i \frac{\omega \mu}{k_c^2} \frac{\partial H_z}{\partial y} = i \frac{\omega \mu k_y}{k_c^2} [C_1 \cos(k_x x) + D_1 \sin(k_x x)] [-C_2 \sin(k_y y) + D_2 \cos(k_y y)] \\ E_y(x, y) = -i \frac{\omega \mu}{k_c^2} \frac{\partial H_z}{\partial x} = -i \frac{\omega \mu k_x}{k_c^2} [-C_1 \sin(k_x x) + D_1 \cos(k_x x)] [C_2 \cos(k_y y) + D_2 \sin(k_y y)] \end{cases}$$

$$\text{边界条件: } \begin{cases} E_x(x, y)|_{y=0} = 0, & E_y(x, y)|_{x=0} = 0 \\ E_y(x, y)|_{x=a} = 0, & E_x(x, y)|_{y=b} = 0 \end{cases}$$

$$\hookrightarrow E_x(x, y) = -i H_0 \frac{\omega \mu k_y}{k_c^2} \cos(k_{x,m} x) \sin(k_{y,n} y)$$

$$E_y(x, y) = i H_0 \frac{\omega \mu k_x}{k_c^2} \sin(k_{x,m} x) \cos(k_{y,n} y)$$

$$k_{x,m} = \frac{m\pi}{a}, \quad k_{y,n} = \frac{n\pi}{b}$$

$$k_z^2 = k_{x,m}^2 + k_{y,n}^2, \quad k_z = \sqrt{k^2 - k_{x,m}^2 - k_{y,n}^2} = \sqrt{\omega^2/\mu\epsilon - k_{x,m}^2 - k_{y,n}^2}$$

类似地:

$$\begin{cases} H_x(x,y) = -iH_0 \frac{k_{x,m} k_z}{k_z^2} \sin(k_{x,m} x) \cos(k_{y,n} y) \\ H_y(x,y) = iH_0 \frac{k_{y,n} k_z}{k_z^2} \cos(k_{x,m} x) \sin(k_{y,n} y) \end{cases}$$

$$H_z(x,y) = H_0 \cos(k_{x,m} x) \cos(k_{y,n} y)$$

m, n 不能同时为 0 (否则 $H_z = 0$)

基模: TE_{10} / TE_{01}

4. 矩形波导管中的 TM 导模

$$E_z \neq 0, \quad H_z = 0$$

$$\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right] E_z(x,y) + k_z^2 E_z(x,y) = 0$$

$$\text{边界条件: } \begin{cases} E_x(x,y)|_{y=0} = 0, \quad E_x(x,y)|_{y=b} = 0 \\ E_y(x,y)|_{x=0} = 0, \quad E_y(x,y)|_{x=a} = 0 \end{cases}$$

$$\text{解得 } E_z(x,y) = E_0 \sin(k_{x,m} x) \sin(k_{y,n} y)$$

$$\begin{cases} E_x(x,y) = iE_0 \frac{k_z k_{x,m}}{k_z^2} \cos(k_{x,m} x) \sin(k_{y,n} y) \\ E_y(x,y) = -iE_0 \frac{k_z k_{y,n}}{k_z^2} \sin(k_{x,m} x) \cos(k_{y,n} y) \\ H_x(x,y) = -iE_0 \frac{\omega\epsilon k_{y,n}}{k_z^2} \sin(k_{x,m} x) \cos(k_{y,n} y) \\ H_y(x,y) = iE_0 \frac{\omega\epsilon k_{x,m}}{k_z^2} \cos(k_{x,m} x) \sin(k_{y,n} y) \end{cases}$$

$m \neq n \neq 0$ (否则 $E_z = 0$)

基模为 TM_{11}

5. 截止频率和单模频段

$$k_z^2 \geq 0 \rightarrow k_{cut,mn} = \sqrt{k_{x,m}^2 + k_{y,n}^2} = \pi \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$$\lambda_{cut,mn} = \frac{2\pi}{k_{cut,mn}}$$

$$f = \frac{\omega}{2\pi} = \frac{k_{cut,mn} c}{2\pi} =$$

截止频率 ω_c 截止波长 λ_c
 简并导模: $k_z = \sqrt{k^2 - k_{min}^2}$ 相同. e.g. TE_{mn} & TM_{mn}

单模区: $\lambda_{cut 0,1} < \lambda < \lambda_{cut 1,0}$

6. 导模波长、相速度及群速度

导模波长 $\lambda_{guiding} = \frac{2\pi}{k_z} = \frac{2\pi}{\sqrt{\mu\epsilon} \sqrt{\omega^2 - \omega_{cut,min}^2}} = \frac{\lambda}{\sqrt{1 - (\frac{\omega_{cut,min}}{\omega})^2}}$ $\lambda = cT = \frac{1}{f} \frac{2\pi}{\omega}$

相速度 $v_p = \frac{\omega}{k_z} = \frac{c}{\sqrt{1 - (\frac{\omega_{cut,min}}{\omega})^2}}$

群速度 $v_g = \frac{d\omega}{dk_z} = \frac{c}{n} \cdot \frac{1}{\sqrt{1 - (\frac{\omega_{cut,min}}{\omega})^2}} = \frac{k_z}{\omega \mu \epsilon}$

$v_p v_g = \frac{1}{\mu\epsilon} = \frac{c^2}{n^2}$

7. 能量密度及传输功率

考虑 TE_m 导模

$$\begin{cases} E_x = E_z = H_y = 0 \\ E_y(\vec{r}, t) = \frac{1}{2} a \omega \mu H_0 \sin(\frac{\pi x}{a}) e^{i(k_z z - \omega t)} \\ H_x(\vec{r}, t) = -\frac{1}{\pi} a k_z H_0 \sin(\frac{\pi x}{a}) e^{i(k_z z - \omega t)} \\ H_z(\vec{r}, t) = H_0 \cos(\frac{\pi x}{a}) e^{i(k_z z - \omega t)} \end{cases} \quad k_z = \sqrt{k^2 - (\frac{\pi}{a})^2}$$

= 0 相位差为 $\frac{\pi}{2}$

$$\begin{aligned} \langle \vec{S} \rangle &= \langle -\text{Re}(E_y) \cdot \text{Re}(H_x) \vec{e}_z + \text{Re}(E_y) \text{Re}(H_z) \vec{e}_x \rangle \\ &= \langle (\frac{a}{2})^2 \omega \mu k_z H_0^2 \sin^2(\frac{\pi x}{a}) \sin^2(k_z z - \omega t) \vec{e}_z \rangle \\ &= \frac{1}{2} (\frac{a}{2})^2 \omega \mu k_z H_0^2 \sin^2(\frac{\pi x}{a}) \vec{e}_z \end{aligned}$$

$$\begin{aligned} P &= \int \langle \vec{S} \rangle \cdot d\vec{S} = \frac{1}{2} (\frac{a}{2})^2 \omega \mu k_z H_0^2 \int_0^b dy \int_0^a dx \sin^2(\frac{\pi x}{a}) \\ &= \frac{1}{4\pi^2} \omega \mu k_z H_0^2 b a^3 \end{aligned}$$

电场振幅 $E_{y0} = \frac{1}{2} a \omega \mu H_0$

$$\begin{aligned} P &= \frac{1}{4} b a \frac{k_z}{\omega \mu} E_{y0}^2 = \frac{1}{4} a b \frac{k}{\omega \mu} \sqrt{1 - (\frac{k_z}{k})^2} E_{y0}^2 \\ &= \frac{1}{4} a b \sqrt{\mu} \sqrt{1 - (\frac{\omega_{cut,min}}{\omega})^2} \end{aligned}$$

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{\lambda} \frac{c}{n} = k \frac{c}{n} = \frac{k}{\epsilon \mu}$$

$$P \propto E_y^2, \quad \omega \rightarrow \omega_{cut} \text{ 时 } P \rightarrow 0$$

工作时 ω 不能太高 (单模区), 也不能过低 ($\omega > \omega_{cut}$)

能量密度:

$$\begin{aligned} w(\vec{r}, t) &= \frac{1}{2} \epsilon \operatorname{Re}(\vec{E}) \cdot \operatorname{Re}(\vec{E}) + \frac{1}{2} \mu \operatorname{Re}(\vec{H}) \cdot \operatorname{Re}(\vec{H}) \\ &= \frac{1}{2} \epsilon \operatorname{Re}^2(E_y) + \frac{1}{2} \mu \left[\operatorname{Re}^2(H_x) + \operatorname{Re}^2(H_z) \right] \\ &= \frac{1}{2} \epsilon \left(\frac{1}{\lambda} a \omega \mu \right)^2 H_0^2 \sin^2\left(\frac{\lambda x}{a}\right) \sin^2(k_z z - \omega t) + \frac{1}{2} \mu \left[\left(\frac{1}{\lambda} a k_z \right)^2 H_0^2 \sin^2\left(\frac{\lambda x}{a}\right) \sin^2(k_z z - \omega t) \right. \\ &\quad \left. + H_0^2 \cos^2\left(\frac{\lambda x}{a}\right) \cos^2(k_z z - \omega t) \right] \\ \langle w \rangle &= \frac{1}{4\pi^2} a \mu H_0^2 \left[\langle \epsilon \mu \omega^2 + k_z^2 \rangle \sin^2\left(\frac{\lambda x}{a}\right) + \left(\frac{\lambda}{a}\right)^2 \cos^2\left(\frac{\lambda x}{a}\right) \right] \end{aligned}$$

$$\begin{aligned} W &= \int \langle w \rangle dS \\ &= \frac{1}{4} \left(\frac{a}{\lambda} \right)^2 \mu H_0^2 \int_0^b dy \int_0^a dx \left[\langle \epsilon \mu \omega^2 + k_z^2 \rangle \sin^2\left(\frac{\lambda x}{a}\right) + \left(\frac{\lambda}{a}\right)^2 \cos^2\left(\frac{\lambda x}{a}\right) \right] \\ &= \frac{1}{4} \left(\frac{a}{\lambda} \right)^2 b \mu H_0^2 = \frac{1}{4} a b \epsilon E_y^2 \end{aligned}$$

$$\frac{P}{W} = \frac{1}{\omega \mu \epsilon} = v_g, \quad \text{群速度: 能量传播速度.}$$

8. 矩形波导管壁面电流分布

考虑 TE₁₀ 导模

$$\vec{K}_f = \vec{n} \times \vec{H}$$

$$\vec{K}_f|_{x=0} = -\vec{e}_x \times \vec{H} = H_0 \cos\left(\frac{\lambda x}{a}\right) \cos(k_z z - \omega t)|_{x=0} = H_0 \cos(k_z z - \omega t)$$

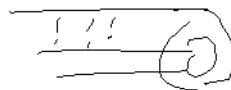
$$\vec{K}_f|_{x=a} = \vec{e}_x \times \vec{H} = -H_0 \cos\left(\frac{\lambda x}{a}\right) \cos(k_z z - \omega t)|_{x=a} = H_0 \cos(k_z z - \omega t)$$

$$\vec{K}_f|_{y=0} = -\vec{e}_y \times \vec{H} = -\frac{1}{\lambda} a k_z H_0 \sin\left(\frac{\lambda x}{a}\right) \sin(k_z z - \omega t) \vec{e}_z + H_0 \cos\left(\frac{\lambda x}{a}\right) \cos(k_z z - \omega t) \vec{e}_x$$

$$\vec{K}_f|_{y=b} = \vec{e}_y \times \vec{H} = \frac{1}{\lambda} a k_z H_0 \sin\left(\frac{\lambda x}{a}\right) \sin(k_z z - \omega t) \vec{e}_z - H_0 \cos\left(\frac{\lambda x}{a}\right) \cos(k_z z - \omega t) \vec{e}_x$$

9. 圆柱内同轴波导管

TE₁₁ 导模:



$$\vec{E}(\vec{r}, t) = \vec{E}(r, \phi) e^{i(k_z z - \omega t)}$$

$$\vec{B}(\vec{r}, t) = \vec{B}(r, \phi) e^{i(k_z z - \omega t)}$$

$$E_z = B_z = 0, \quad k = k_z$$

$$\nabla \times \vec{E} = (\nabla \times \vec{E})_z = -\frac{\partial B_z}{\partial t} = 0 \rightarrow \vec{E}(r, \phi) = -\nabla \varphi(r, \phi)$$

$$\text{Laplace 方程: } \nabla^2 \varphi(r, \phi) = 0$$

$$\text{轴对称解为 } \varphi(r) = a_0 + b_0 \ln r$$

$$\vec{E}(r) = -\nabla \varphi(r) = -b_0 \frac{\vec{r}}{r^2}$$

$$\vec{E}(\vec{r}, t) = \vec{E}_0 e^{i(kz - \omega t)} \frac{1}{r} \vec{e}_r$$

$$\nabla \times \vec{E} = -\frac{\partial}{\partial t} (\vec{B}) \quad i k E_r = i \omega B_\phi$$

$$\vec{B}(\vec{r}, t) = \frac{k}{\omega} \vec{E}_0 e^{i(kz - \omega t)} = \frac{1}{\mu_0 c} \vec{E}_0 e^{i(kz - \omega t)} \frac{1}{r} \vec{e}_\phi$$

$$\vec{S} = \text{Re}(\vec{E}) \times \frac{1}{\mu_0} \text{Re}(\vec{B})$$

$$= \frac{\sqrt{\epsilon}}{\mu_0} E_0^2 \frac{1}{r^2} \cos^2(kz - \omega t) \vec{e}_z$$

$$\langle \vec{S} \rangle = \frac{1}{2} \frac{\sqrt{\epsilon}}{\mu_0} \frac{E_0^2}{r^2} \vec{e}_z$$

$$P = \int \langle \vec{S} \rangle \cdot d\vec{s} = \frac{1}{2} \frac{\sqrt{\epsilon}}{\mu_0} E_0^2 \int_{r_1}^{r_2} \frac{1}{r^2} 2\pi r dr = \pi \frac{\sqrt{\epsilon}}{\mu_0} E_0^2 \ln \frac{r_2}{r_1}$$

$$\vec{K}_f|_{r=r_1} = \vec{e}_r \times \vec{H} = \frac{\sqrt{\epsilon}}{\mu_0} E_0 \frac{1}{r_1} \cos(kz - \omega t) \vec{e}_\phi$$

$$\vec{K}_f|_{r=r_2} = -\vec{e}_r \times \vec{H} = -\frac{\sqrt{\epsilon}}{\mu_0} E_0 \frac{1}{r_2} \cos(kz - \omega t) \vec{e}_\phi$$

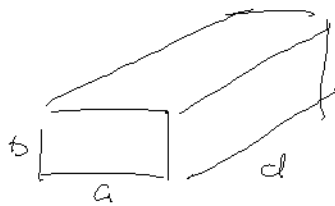
$$G_f|_{r=r_1} = \vec{e}_r \cdot \text{Re}(\epsilon \vec{E})|_{r=r_1} = \epsilon E_0 \frac{1}{r_1} \cos(kz - \omega t)$$

$$G_f|_{r=r_2} = -\vec{e}_r \cdot \text{Re}(\epsilon \vec{E})|_{r=r_2} = -\epsilon E_0 \frac{1}{r_2} \cos(kz - \omega t)$$

10. 金属导体谐振腔中的TE/TM共振模。

1) TE 导模

$$H_z(\vec{r}, t) = h_0 \cos(k_{x,m} x) \cos(k_{y,n} y) e^{i(k_z z - \omega t)}$$



边界条件: $H_z|_{z=0} = 0$, $H_z|_{z=d} = 0$ ($k_y = 0$)

设 $H_z(\vec{r}, t) = \cos(k_{x,m} x) \cos(k_{y,n} y) [h_0 e^{i(k_z z - \omega t)} - h_0' e^{i(-k_z z - \omega t)}]$

$$\begin{cases} h_0 + h_0' = 0 & \text{解得 } h_0 = -h_0' \\ \sin(k_z d) = 0 & k_z = k_{z,l} = \frac{l\pi}{d} \end{cases}$$

驻波解: $H_z = H_0 \cos(k_{x,m} x) \cos(k_{y,n} y) \sin(k_{z,l} z) e^{-i\omega t}$ $H_0 = 2ih_0$

$$E_x = (-i \frac{\omega \mu k_{y,n}}{k^2}) H_0 \cos(k_{x,m} x) \sin(k_{y,n} y) \sin(k_{z,l} z) e^{-i\omega t}$$

$$E_y = i \frac{\omega \mu k_{x,m}}{k^2} H_0 \sin(k_{x,m} x) \cos(k_{y,n} y) \sin(k_{z,l} z) e^{-i\omega t}$$

$$H_x = -\frac{k_{x,m} k_{z,l}}{k^2} H_0 \sin(k_{x,m} x) \cos(k_{y,n} y) \cos(k_{z,l} z) e^{-i\omega t}$$

$$H_y = -\frac{k_{y,n} k_{z,l}}{k^2} H_0 \cos(k_{x,m} x) \sin(k_{y,n} y) \cos(k_{z,l} z) e^{-i\omega t}$$

$$k^2 = k_{x,m}^2 + k_{y,n}^2 = k^2 - k_{z,l}^2$$

$$\text{共振波数 } k_{mnl} = 2\pi \sqrt{(\frac{m}{a})^2 + (\frac{n}{b})^2 + (\frac{l}{d})^2}$$

m, n 不同时为 0, l 不为 0 (否则 $\sin(k_{z,l} z) = 0$, $E = 0$)

若 m, n

为模 $(1, 0, 1)$ $(0, 1, 1)$

(2) TM 模

$$E_z \neq 0, H_z = 0$$

$$E_z = E_0 \sin(k_{x,m} x) \sin(k_{y,m} y) \cos(k_{z,1} z) e^{-i\omega t}$$

$$E_x = -\frac{k_{z,1} k_{x,m}}{k^2} E_0 \cos(k_{x,m} x) \sin(k_{y,m} y) \sin(k_{z,1} z) e^{-i\omega t}$$

$$E_y = -\frac{k_{z,1} k_{y,m}}{k^2} E_0 \sin(k_{x,m} x) \cos(k_{y,m} y) \sin(k_{z,1} z) e^{-i\omega t}$$

$$H_x = -i \frac{\omega \epsilon k_{y,m}}{k^2} E_0 \sin(k_{x,m} x) \cos(k_{y,m} y) \cos(k_{z,1} z) e^{-i\omega t}$$

$$H_y = i \frac{\omega \epsilon k_{x,m}}{k^2} E_0 \cos(k_{x,m} x) \sin(k_{y,m} y) \cos(k_{z,1} z) e^{-i\omega t}$$

模 $(1, 1, 0)$

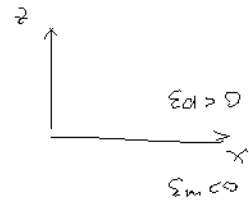
11. 电磁波沿介电常数异号的界面的传播.

$\omega < \omega_p$ 时, 金属 $\text{Re}(\epsilon) < 0$

不能在 $\text{Re}(\epsilon) < 0$ 的物体内传播, 但可在表面传播.

(1) TM 模:

$$\vec{H} = \begin{cases} A e^{i(k_x x + k_{d,z} z - \omega t)} \vec{e}_y & (z > 0) \\ B e^{i(k_x x + k_{m,z} z - \omega t)} \vec{e}_y & (z < 0) \end{cases}$$



$$\begin{cases} \epsilon_d k^2 = k_x^2 + k_{d,z}^2 & k^2 = \epsilon k_0^2 = k_x^2 + k_z^2 \\ \epsilon_m k^2 = k_x^2 + k_{m,z}^2 \end{cases}$$

$$\nabla \times \vec{H} = \epsilon \frac{\partial \vec{E}}{\partial t}, \quad \epsilon = \epsilon_0 \epsilon_d \quad (z > 0), \quad \epsilon_0 \epsilon_m \quad (z < 0)$$

$$\vec{E} = \begin{cases} \frac{-1}{\omega \epsilon_0 \epsilon_d} [k_x \vec{e}_z - k_{d,z} \vec{e}_x] A e^{i(k_x x + k_{d,z} z - \omega t)} & (z > 0) \\ \frac{-1}{\omega \epsilon_0 \epsilon_m} [k_x \vec{e}_z - k_{m,z} \vec{e}_x] B e^{i(k_x x + k_{d,z} z - \omega t)} & (z < 0) \end{cases}$$

边界关系: $\begin{cases} H_y|_{z=0^+} = H_y|_{z=0^-} & H, E \text{ 切向分量连续.} \\ E_x|_{z=0^+} = E_x|_{z=0^-} \end{cases}$

$$\Rightarrow A = B, \quad \frac{A k_{d,z}}{\epsilon_d} = \frac{B k_{m,z}}{\epsilon_m}$$

$$\text{解得 } k_x = k_0 \sqrt{\frac{\epsilon_d \epsilon_m}{\epsilon_d + \epsilon_m}}, \quad k_{dz} = \frac{\epsilon_d k_0}{\epsilon_d + \epsilon_m}, \quad k_{mz} = \frac{\epsilon_m k_0}{\epsilon_d + \epsilon_m}$$

$$\epsilon_m < 0 :$$

$|\epsilon_m| < \epsilon_d$ 时, k_x 为实数, k_{dz} 、 k_{mz} 为虚数,

⇒ 具有表面波特性, 在界面两侧衰减

SPP (Surface Plasmon Polarization)

2) TE 模:

$$\vec{E} = \begin{cases} A e^{i(k_x x + k_{dz} z - \omega t)} \vec{e}_y & (z > 0) \\ B e^{i(k_x x + k_{mz} z - \omega t)} \vec{e}_y & (z < 0) \end{cases}$$

$$\nabla \times \vec{E} = -\mu_0 \frac{\partial \vec{H}}{\partial t}$$

$$\vec{H} = \begin{cases} \frac{A}{\mu_0 \omega} (k_x \vec{e}_z - k_{dz} \vec{e}_x) e^{i(k_x x + k_{dz} z - \omega t)} \vec{e}_y \\ \frac{B}{\mu_0 \omega} (k_x \vec{e}_z - k_{mz} \vec{e}_x) e^{i(k_x x + k_{mz} z - \omega t)} \vec{e}_y \end{cases}$$

12. SPP 的色散特性.

$$\text{金属导体: } \epsilon(\omega) = 1 - \frac{\omega_p^2}{\omega^2 + i\delta\omega}$$

$$\epsilon_m(\omega) = 1 - \frac{\omega_p^2}{\omega^2} \quad (\delta = 0)$$

13. SPP 的传播距离

$$\epsilon_m = \epsilon_m' + i\epsilon_m'', \quad k_x = k_x' + i k_x''$$

$$= \frac{\omega}{c} \sqrt{\frac{\epsilon_m' \epsilon_d}{\epsilon_m' + \epsilon_d}} + i \frac{\omega}{c} \left(\sqrt{\frac{\epsilon_m' \epsilon_d}{\epsilon_m' + \epsilon_d}} \right)^3 \frac{\epsilon_m''}{(\epsilon_m')^2}$$

$$E_{SPP} \propto e^{i(k_x x - \omega t)} = e^{-k_x'' x} e^{i(k_x' x - \omega t)}$$

$$\lambda_{SPP} = \frac{2\pi}{k_x'}$$

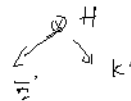
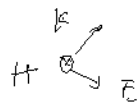
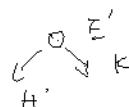
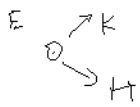
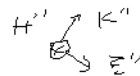
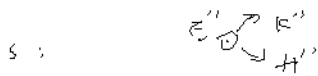
$$I_{SPP} = |E_{SPP}|^2 \propto e^{-2k_x'' x}$$

$$L_{SPP} = \frac{1}{2k''}$$

复习：

一、1. Fresnel 公式：

s 偏振：E ⊥ 入射面、p 偏振：E // 入射面。



边界条件

$$\begin{cases} E_t + E_r = E_i \\ H_t + H_r = H_i \end{cases}$$

① s 偏振

$$\begin{cases} E + E' = E'' \\ H \cos \theta - H' \cos \theta' = H'' \cos \theta'' \end{cases}$$

$$\theta' = \theta, \quad H = \frac{E}{Z}$$

$$\begin{cases} E + E' = E'' \\ \frac{1}{z_1} (E - E') \cos \theta = \frac{1}{z_2} E'' \cos \theta'' \end{cases}$$

$$\text{解得 } \left(\frac{E'}{E}\right)_s = \frac{z_2 \cos \theta - z_1 \cos \theta''}{z_2 \cos \theta + z_1 \cos \theta''} = \left(\frac{H'}{H}\right)_s$$

$$\left(\frac{E''}{E}\right)_s = \frac{2z_2 \cos \theta}{z_2 \cos \theta + z_1 \cos \theta''} = \frac{z_2}{z_1} \left(\frac{H''}{H}\right)_s$$

② p 偏振:

$$\begin{cases} (E - E') \cos \theta = E'' \cos \theta'' \\ H + H' = H'' \end{cases}$$

$$\text{解得 } \left(\frac{E'}{E}\right)_p = \frac{z_1 \cos \theta - z_2 \cos \theta''}{z_1 \cos \theta + z_2 \cos \theta''} = \left(\frac{H'}{H}\right)_p$$

$$\left(\frac{E''}{E}\right)_p = \frac{2z_2 \cos \theta}{z_1 \cos \theta + z_2 \cos \theta''} = \frac{z_2}{z_1} \left(\frac{H''}{H}\right)_p$$

垂直入射时: $\frac{E'}{E} = \frac{z_2 - z_1}{z_2 + z_1}$, $\frac{E''}{E} = \frac{2z_2}{z_2 + z_1}$

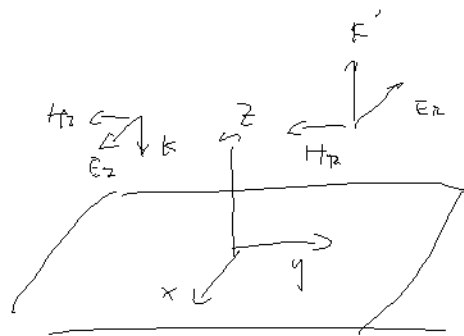
若 $z_1 = z_2$, 则无反射 — 阻抗匹配.

辐射场求解:

$$E_z(t) = E_{z0} e^{-i(k_0 z + \omega t)}$$

$$\begin{aligned} \vec{H}|_{z=0} &= -(H_{T0} + H_{R0}) e^{-i\omega t} \vec{e}_y \\ &= -\frac{2}{z_1} E_{z0} e^{-i\omega t} \vec{e}_y \end{aligned}$$

$$\begin{aligned} \vec{K}_p &= \vec{n} \times \vec{H} = \vec{e}_z \times \vec{H}|_{z=0} \\ &= \frac{2}{z_1} E_{z0} e^{-i\omega t} \vec{e}_x \end{aligned}$$



2. Brewster 角: 对非磁性介质 $\epsilon = \bar{\mu} = 1$

$$\left(\frac{E'}{E}\right)_s = \frac{n_1 \cos \theta - n_2 \cos \theta''}{n_1 \cos \theta + n_2 \cos \theta''}$$

$$\left(\frac{E'}{E}\right)_p = \frac{n_2 \cos \theta - n_1 \cos \theta''}{n_2 \cos \theta + n_1 \cos \theta''}$$

$$n_1 \sin \theta = n_2 \sin \theta'$$

$$E'_s = 0 \rightarrow \cos \theta = \frac{n_2}{n_1} \cos \theta'$$

$$\Rightarrow \left(\frac{n_1}{n_2}\right)^2 = 1, \text{ 矛盾.}$$

$$E'_p = 0 \rightarrow \cos \theta = \frac{n_1}{n_2} \cos \theta'$$

$$\left(\frac{n_1}{n_2}\right)^2 \sin^2 \theta + \left(\frac{n_1}{n_1}\right)^2 \cos^2 \theta = 1 \quad \left(\frac{n_1}{n_2}\right)^2 \tan^2 \theta + \left(\frac{n_2}{n_1}\right)^2 = 1 + \tan^2 \theta$$

$$\text{解得} \tan^2 \theta = \frac{1 - \left(\frac{n_2}{n_1}\right)^2}{\left(\frac{n_1}{n_2}\right)^2 - 1} = \left(\frac{n_2}{n_1}\right)^2$$

$$\Rightarrow \theta_B = \arctan\left(\frac{n_2}{n_1}\right)$$

3. 全反射: $n_1 > n_2$, $\theta = \theta_c$ 时 $(\theta_c = \arcsin(\frac{n_2}{n_1}))$ 此时 $\theta' = \frac{\pi}{2}$

$$\left(\frac{E'}{E}\right)_s = \frac{n_1 \cos \theta - n_2 \cos \theta'}{n_1 \cos \theta + n_2 \cos \theta'} = \frac{n_1 \cos \theta_c}{n_1 \cos \theta_c}$$

$$\left(\frac{E'}{E}\right)_p = \frac{n_2 \cos \theta_c}{n_2 \cos \theta_c} = 1 \quad \text{其实方向(含义)与入射光相反}$$

$\Rightarrow E' = 0$ 无反射

波数:

$$k_2 = \frac{\omega}{c} \cdot n_2 = \frac{n_2}{n_1} k_1$$

$$k_{2x} = k_2 \sin \theta' = \frac{n_1}{n_2} \sin \theta k_2 = k_1 \sin \theta$$

$$\begin{aligned} k_{2z} &= \sqrt{k_2^2 - k_{2x}^2} \\ &= k_1 \sqrt{\left(\frac{n_2}{n_1}\right)^2 - \sin^2 \theta} = i k_1 \sqrt{\sin^2 \theta - \sin^2 \theta_c} \\ &= i k \end{aligned}$$

$$\vec{E}'' = \vec{E}' e^{i[(ik_z + k_{2x}x) - \omega t]}$$

$$= \vec{E}'' e^{-ikz} e^{i(k_2 x - \omega t)}$$

curl: $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \rightarrow i\vec{k} \times \vec{E} = i\omega \vec{B} \rightarrow \vec{B} = \frac{1}{\omega} (\vec{k} \times \vec{E})$

$$\vec{E}'' = E'' \vec{e}_y$$

$$B_{yz} = \frac{1}{\omega} k_{yz} E'' \quad B_{yx} = \frac{1}{\omega} k_{yz} E'' = i \frac{1}{\omega} k E''$$

$$\langle S_{yx} \rangle \neq 0, \quad \langle S_{yz} \rangle = 0$$

二、介质与金属色散.

1. Lorentz 模型.

$$m \frac{d^2 \vec{x}}{dt^2} = -m\omega_0^2 \vec{x} - \gamma \frac{d\vec{x}}{dt} + (-e \vec{E}_0 e^{-i\omega t}) \quad \vec{x} = \vec{x}_0 e^{-i\omega t}$$

求解 $\vec{x} = -\frac{e\vec{E}_0}{m} \frac{1}{\omega_0^2 - \omega^2 - i\gamma\omega} e^{-i\omega t}$

$$\begin{aligned} \text{极化强度 } \vec{P} &= \vec{p} = -Ne\vec{x} \\ &= \frac{Ne^2}{m} \frac{1}{\omega_0^2 - \omega^2 - i\gamma\omega} \vec{E} \end{aligned}$$

极化率, $\chi = \frac{\vec{P}}{\epsilon_0 \vec{E}} = \frac{Ne^2/\epsilon_0 m}{\omega_0^2 - \omega^2 - i\gamma\omega}$

则 $\omega_p = \sqrt{\frac{Ne^2}{\epsilon_0 m}}$ 为等离子体频率

$$\chi = \frac{\omega_p^2}{\omega_0^2 - \omega^2 - i\gamma\omega}$$

$$\chi = \frac{\omega_p^2}{\omega_0^2 - \omega^2 - i\gamma\omega}$$

$$= 1 + \frac{\omega_p^2 (\omega_0^2 - \omega^2)}{(\omega_0^2 - \omega^2)^2 + (\gamma \omega)^2} + i \frac{\omega_p^2 \gamma \omega}{(\omega_0^2 - \omega^2)^2 + (\gamma \omega)^2}$$

$$= \epsilon_r' + i \epsilon_r''$$

其中 $\epsilon_r' = 1 + \frac{\omega_p^2 (\omega_0^2 - \omega^2)}{(\omega_0^2 - \omega^2)^2 + (\gamma \omega)^2}$

$$\epsilon_r'' = \frac{\omega_p^2 \gamma \omega}{(\omega_0^2 - \omega^2)^2 + (\gamma \omega)^2}$$

$\omega = \omega_0$: 共振, 此时 $\text{Im}(\epsilon_r)$ 极大, 表现为强烈吸收效应。

2. Drude 模型.

$$m \frac{d\vec{v}}{dt} = -e \vec{E} + (-e \vec{E}_0 e^{-i\omega t})$$

解得 $\vec{v} = \frac{e \vec{E}_0}{m} \frac{1}{\omega^2 + i\gamma \omega} e^{-i\omega t}$

$$\vec{J} = -Ne \vec{v} = -Ne \frac{d\vec{x}}{dt} = i\omega Ne \vec{x}$$

$$\sigma = \frac{\vec{J}}{\vec{E}} = i \frac{Ne^2}{\epsilon_0 m} \frac{\epsilon_0 \omega}{\omega^2 + i\gamma \omega} = \frac{\epsilon_0 \omega p^2}{\gamma - i\omega}$$

直流电阻: $\omega \rightarrow 0, \sigma = \frac{\epsilon_0 \omega p^2}{\gamma} = \frac{Ne^2}{\gamma m}$

$$\epsilon_{\text{opt}} = \epsilon_b + \frac{i\sigma}{\omega} \rightarrow \epsilon_b - \frac{1}{\omega} \text{Im}(\sigma) + i \cdot \frac{1}{\omega} \text{Re}(\sigma)$$

$$= \epsilon_b - \frac{1}{\omega} \frac{\epsilon_0 \omega p^2 \cdot \omega}{\gamma^2 + \omega^2} + i \frac{1}{\omega} \cdot \frac{\epsilon_0 \omega p^2 \cdot \gamma}{\gamma^2 + \omega^2}$$

$$= (\epsilon_r' + i \epsilon_r'') \epsilon_0 \quad \epsilon_b = \epsilon_0$$

其中 $\epsilon_r' = 1 - \frac{\omega_p^2}{\gamma^2 + \omega^2}$

$$\epsilon_r'' = \frac{\omega_p^2 \gamma}{\omega (\gamma^2 + \omega^2)}$$

3. 单色平面电磁波在导电媒质中的传播.

$$\left\{ \begin{array}{l} \nabla \cdot \vec{E} = 0 \quad (\rho_f = 0) \\ \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \\ \nabla \cdot \vec{B} = 0 \\ \nabla \times \vec{B} = \mu_0 (\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t}) = \mu_0 (\sigma \vec{E} - i\omega \epsilon_0 \vec{E}) \\ = -i(-i\omega) (\epsilon_b + \frac{i\sigma}{\omega}) \vec{E} \end{array} \right.$$

$$= \mu_0 \epsilon_{eff} \frac{\partial \vec{E}}{\partial t}$$

$$\vec{E} = \vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)}, \quad \vec{B} = \vec{B}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)}$$

$$\vec{k} \cdot \vec{E}_0 = 0, \quad \vec{k} \cdot \vec{B}_0 = 0$$

$$\vec{k} \times \vec{E}_0 = -\omega \vec{B}_0$$

$$\vec{k} \times \vec{B}_0 = -\mu_0 \epsilon_{eff} \omega \vec{E}_0$$

色散关系, $k = \omega \sqrt{\mu_0 \epsilon_{eff}(\omega)}$

设 $\vec{k} = \vec{\alpha} + i\vec{\beta}$

$$|\vec{k}|^2 = \vec{k} \cdot \vec{k}$$